

Revisit for Module and Representation Theory

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1 Concepts

1. faithful, injective, projective, flat, Noetherian, semisimple, Artinian, Soc, essential epimorphism, Jacobson radical, projective cover, DVR.

Composition series, Wedderburn's Theorem, idempotents, Schur's lemma, the cartan matrix, Krull-Schmidt, Socle Series, Radical Series, Top, Nakayama lemma, Idempotent Lifting Jordan-Hölder, Loewy

2 Module

Theorem 2.1. *The following three conditions on V_A are equivalent.*

- (1) V is a Noetherian module.
- (2) (Ascending chain condition.) For every infinite chain of A -submodules of V ,

$$V_1 \subset V_2 \subset \cdots \subset V_n \subset \cdots, \quad (2.1)$$

there is a number m such that $V_m = V_{m+1} = \cdots$.

- (3) Every A -submodule of V is finitely generated over A .

Remark 2.2 (Finiteness in Category Theory). **(1) Finite Presentation via Colimits:**

In classical module categories, “finiteness” is characterized by filtered colimits. A module M is finitely presented if and only if the functor $\text{Hom}(M, -)$ commutes with filtered colimits:

$$\varinjlim \text{Hom}(M, N_i) \xrightarrow{\sim} \text{Hom}(M, \varinjlim N_i)$$

Intuitively, any morphism from M to an infinite filtered system must factor through a finite stage.

(2) Finiteness via Functor Preservation in ∞ -Categories:

In higher category theory, objects lack “elements.” Instead, an object X is defined as “finite” based on how its associated functors interact with limits and colimits.

- **Compact (Finitely Presented) Objects:**

The functor $\text{Map}(X, -)$ commutes with *filtered colimits*.

Example: Finite CW complexes in the ∞ -category of Spaces.

- **Dualizable Objects:**

The functor $X \otimes -$ commutes with *all limits* (e.g., infinite products). This happens because having a dual allows the left adjoint $X \otimes -$ to act like the right adjoint $\underline{\text{Map}}(X^\vee, -)$.

Example: Finite spectra in the stable ∞ -category of Spectra.

- **Perfect Complexes:**

The ultimate intersection of both finiteness conditions. $\mathbb{R}\text{Map}(X, -)$ preserves filtered colimits, and $X \otimes^{\mathbb{L}} -$ preserves limits.

Example: Perfect complexes in the derived ∞ -category $\mathcal{D}(R)$ or $\text{QCoh}(X)$ in derived algebraic geometry.

Theorem 2.3. *The following three conditions on a module V_A are equivalent:*

- (1) V is an Artinian module.
- (2) (Descending chain condition.) For every infinite chain of A -submodules of V ,

$$V_1 \supset V_2 \supset \cdots \supset V_n \supset \cdots,$$

there is a number m such that $V_m = V_{m+1} = \cdots$.

(3) (Minimal condition.) Every non-empty set of A -submodules of V contains a minimal element with respect to inclusion.

Theorem 2.4. Let W be an A -submodule of V . Then V is Noetherian (respectively, Artinian) if and only if both W and V/W are Noetherian (respectively, Artinian).

Definition 2.5. A nonzero A -module V is said to be **irreducible** (or **simple**) if it has no proper A -submodule other than 0.

A chain of A -submodules of V

$$V = V_0 \supset V_1 \supset \cdots \supset V_n = 0$$

is said to be a **composition series** of V if each V_i/V_{i+1} is irreducible. And the simple modules

$$V/V_1, V_1/V_2, \dots, V_{n-1}/V_n = V_{n-1}$$

are called the **composition factors** of the series.

Theorem 2.6. A non-zero module has a composition series if and only if it is both Noetherian and Artinian.

Proof. Use [Theorem 2.4](#) and induction on the length of chain. □

Theorem 2.7 (Jordan–Hölder). If V has a composition series, any two composition factors obtained from the two series of V are the same up to order of occurrence and isomorphism.

Theorem 2.8 (Equivalent descriptions of the Jacobson radical). Let A be a unital associative ring. For $a \in A$, the following conditions are equivalent:

- (i) $a \in J(A)$.
- (ii) a annihilates every simple left A -module, i.e. $aS = 0$ for every simple left A -module S .
- (iii) a annihilates every simple right A -module, i.e. $Ta = 0$ for every simple right A -module T .
- (iv) a belongs to every maximal left ideal of A .
- (v) a belongs to every maximal right ideal of A .
- (vi) $1 - xa \in A^\times$ for every $x \in A$.
- (vii) $1 - ax \in A^\times$ for every $x \in A$.

Equivalently,

$$J(A) = \bigcap_{S \text{ simple}} \text{Ann}_A(S) = \bigcap_{\mathfrak{m} \text{ maximal left}} \mathfrak{m} = \bigcap_{\mathfrak{n} \text{ maximal right}} \mathfrak{n}.$$

Moreover, $J(A)$ is the largest two-sided ideal $I \subset A$ such that $1 - a \in A^\times$ for every $a \in I$.

Theorem 2.9. Let A be a unital associative ring. If $I \subset A$ is a nil left(right, double-sided) ideal, then $I \subset J(A)$.

Proof. Use the (vi) of [Theorem 2.8](#). □

Theorem 2.10 (Jacobson radical of an Artinian ring). Let A be a unital associative ring. If A is left Artinian or right Artinian, then $J(A)$ is the largest nilpotent two-sided ideal of A .

Theorem 2.11. *Let A be a unital associative ring. If S is a simple left A -module, then there exists a maximal left ideal $\mathfrak{m} \subset A$ such that*

$$S \cong A/\mathfrak{m}$$

as left A -modules.

Remark 2.12. Let A be a unital associative ring, and let $\mathfrak{m} \subset A$ be a maximal left ideal. Then $A/\mathfrak{m}$ is a simple left A -module, and

$$\text{Ann}_A(A/\mathfrak{m}) = \{a \in A \mid aA \subset \mathfrak{m}\}.$$

This is the largest two-sided ideal contained in $\mathfrak{m}$. In general it need not equal $\mathfrak{m}$, since $\mathfrak{m}$ is only a left ideal. Thus $\mathfrak{m}$ is the annihilator of the element $1 + \mathfrak{m}$, whereas $\text{Ann}_A(A/\mathfrak{m})$ is the annihilator of the whole module.

Theorem 2.13 (Azumaya–Nakayama). *Let A be a unital associative ring.*

(i) *Let V be a left A -module and let W be an A -submodule of V such that V/W is finitely generated over A . Then*

$$V = W + J(A)V \implies V = W.$$

(ii) *If V is a finitely generated left A -module, then*

$$J(A)V = V \implies V = 0.$$

Theorem 2.14. *Let A be a unital associative ring, and let M be a left A -module of finite length. Then*

$$M \text{ is semisimple} \iff \text{rad}(M) = 0.$$

Theorem 2.15. *Let A be a unital associative ring.*

(i) *If A is semisimple, then $J(A) = 0$.*

(ii) *If A is left Artinian or right Artinian, then*

$$A \text{ is semisimple} \iff J(A) = 0.$$

Definition 2.16. Let A be a unital associative ring, and let M be a left A -module. The radical of M is defined by

$$\text{rad}(M) = \bigcap_{N \subset M \text{ maximal}} N,$$

where the intersection is taken over all maximal submodules of M .

Corollary 2.17. *Let A be a unital associative ring, and let M be a left A -module. Then*

$$J(A)M \subset \text{rad}(M).$$

Theorem 2.18. *Let A be a unital associative ring, and let M be a left A -module. If M is semisimple, then*

$$\text{rad}(M) = 0.$$

Theorem 2.19. *Let A be a left Artinian ring, and let M be a finitely generated left A -module. Then*

$$\text{rad}(M) = J(A)M.$$

Proof. Use [Theorem 2.18](#). □

Theorem 2.20. *Let A be a unital associative ring, and let M be a left A -module. A decomposition*

$$M = M_1 \oplus \cdots \oplus M_n$$

is equivalent to a decomposition of id_M in $\text{End}_A(M)$ as

$$\text{id}_M = p_1 + \cdots + p_n,$$

where $p_i^2 = p_i$ and $p_i p_j = 0$ for $i \neq j$. Under this correspondence,

$$M_i = \text{Im}(p_i).$$

In particular, M is indecomposable if and only if id_M is a primitive idempotent in $\text{End}_A(M)$.

Theorem 2.21. *Let A be a unital associative ring. Then*

$$\text{End}_A(A_A) \cong A \quad \text{End}_A({}_A A) \cong A^{op}.$$

Theorem 2.22. *Let A be a unital associative ring, and let $e \in A$ be an idempotent.*

(i) If $e = e_1 + \cdots + e_n$ is a decomposition into pairwise orthogonal idempotents, then

$$eA = e_1A \oplus \cdots \oplus e_nA$$

as right A -modules, and

$$Ae = Ae_1 \oplus \cdots \oplus Ae_n$$

as left A -modules.

(ii) Conversely, every direct sum decomposition

$$eA = I_1 \oplus \cdots \oplus I_n$$

into right ideals is of the form $I_i = e_iA$, where $e = e_1 + \cdots + e_n$ is a decomposition into pairwise orthogonal idempotents. Similarly, every direct sum decomposition

$$Ae = L_1 \oplus \cdots \oplus L_n$$

into left ideals is of the form $L_i = Ae_i$, where $e = e_1 + \cdots + e_n$ is a decomposition into pairwise orthogonal idempotents.

Theorem 2.23. *Let A be a unital associative ring, let M be a left A -module, and set $E = \text{End}_A(M)$. For a nonzero idempotent $p \in E$, write $pM = \text{Im}(p)$. Then*

$$p \text{ is primitive in } E \iff pM \text{ is indecomposable as a left } A\text{-module.}$$

Moreover,

$$\text{End}_A(pM) \cong pEp.$$

Theorem 2.24. *Let A be a unital associative ring, and let V be a left A -module. Suppose that V has a composition series. Then*

$$V \text{ is indecomposable} \iff \text{End}_A(V) \text{ is a local ring.}$$

Theorem 2.25 (Krull–Schmidt–Azumaya). *Let A be a unital associative ring, and let V be a left A -module. Suppose that*

$$V = V_1 \oplus \cdots \oplus V_n$$

is a finite direct sum decomposition into indecomposable submodules, and that $\text{End}_A(V_i)$ is a local ring for every i .

Then this decomposition is unique up to isomorphism and permutation. That is, if

$$V = W_1 \oplus \cdots \oplus W_m$$

is another finite direct sum decomposition into indecomposable submodules, then $m = n$, and there exists a permutation $\sigma \in S_n$ such that

$$V_i \cong W_{\sigma(i)}$$

for all $1 \leq i \leq n$.

Lemma 2.26. *Let A be a unital associative ring, and let M be an Artinian left A -module. Set $E = \text{End}_A(M)$. Then every nonzero idempotent $p \in E$ admits a finite decomposition*

$$p = p_1 + \cdots + p_n$$

where $p_1, \dots, p_n \in E$ are pairwise orthogonal primitive idempotents. Equivalently,

$$\text{Im}(p) = \text{Im}(p_1) \oplus \cdots \oplus \text{Im}(p_n)$$

is a finite direct sum decomposition into indecomposable left A -modules.

Proof. Since p is idempotent, $\text{Im}(p)$ is a direct summand of M . Hence $\text{Im}(p)$ is Artinian. Repeatedly decompose $\text{Im}(p)$ whenever it is decomposable. This process terminates, because otherwise one obtains an infinite strictly descending chain of submodules of M . Thus

$$\text{Im}(p) = M_1 \oplus \cdots \oplus M_n$$

with each M_i indecomposable.

Let p_i be the projection from M onto M_i , with kernel equal to the sum of the other summands and $\ker(p)$. Then the p_i are pairwise orthogonal idempotents in E , and

$$p = p_1 + \cdots + p_n.$$

Since $\text{Im}(p_i) = M_i$ is indecomposable, p_i is primitive. □

Corollary 2.27. *Let A be a left Artinian ring, and let M be a finitely generated left A -module. Then M admits a finite direct sum decomposition*

$$M = M_1 \oplus \cdots \oplus M_n$$

where each M_i is indecomposable. Moreover, this decomposition is unique up to isomorphism and permutation.

Proof. Use [Lemma 2.26](#) and [Theorem 2.25](#). □

Theorem 2.28. *Let A be a unital associative ring, and let P be a left A -module. The following conditions are equivalent:*

(i) *P is projective.*

(ii) For every epimorphism $M \twoheadrightarrow N$ and every homomorphism $P \rightarrow N$, there exists a lift $P \rightarrow M$.

(iii) The functor

$$\mathrm{Hom}_A(P, -) : A\text{-Mod} \longrightarrow \mathrm{Ab}$$

is exact.

(iv) P is a direct summand of a free left A -module.

(v) Every epimorphism $M \twoheadrightarrow P$ splits.

If P is finitely generated, then these conditions are also equivalent to P being a direct summand of A^n for some $n \geq 1$.

Theorem 2.29. Let A be a unital ring, and let I be a left A -module. The following conditions are equivalent:

(i) I is injective, i.e., for every submodule $M \subset N$ and every homomorphism $f : M \rightarrow I$, there exists an extension $g : N \rightarrow I$ such that $g|_M = f$.

(ii) The functor

$$\mathrm{Hom}_A(-, I) : (A\text{-Mod})^{\mathrm{op}} \longrightarrow \mathrm{Ab}$$

is exact.

(iii) (Baer criterion) For every left ideal $J \subset A$ and homomorphism $f : J \rightarrow I$, there exists an extension $g : A \rightarrow I$ with $g|_J = f$.

Theorem 2.30. Let A be a unital associative ring, and let F be a left A -module. The following conditions are equivalent:

(i) F is flat, i.e. the functor

$$- \otimes_A F : \mathrm{Mod}\text{-}A \longrightarrow \mathrm{Ab}$$

is exact. Equivalently, for every monomorphism $M' \hookrightarrow M$ of right A -modules, the induced map

$$M' \otimes_A F \longrightarrow M \otimes_A F$$

is a monomorphism.

(ii) For every finitely generated right ideal $I \subset A$, the natural map

$$I \otimes_A F \longrightarrow A \otimes_A F$$

is injective.

(iii) F is a filtered colimit of finitely generated free left A -modules.

Theorem 2.31 (Wedderburn–Artin). Let A be a simple Artinian ring. Then there exist a division ring D and a positive integer n such that

$$A \cong M_n(D),$$

the ring of $n \times n$ matrices over D .

Moreover, as a left A -module, the left regular module ${}_A A$ decomposes as

$${}_A A \cong S^{\oplus n},$$

where $S = Ae_{11}$ is a minimal left ideal of A , simple and projective, corresponding to the first column of the matrix block. Each of the n summands is isomorphic to S , so the left regular module is a direct sum of mutually isomorphic minimal left ideals.

Theorem 2.32 (Idempotent lifting modulo a nilpotent ideal). *Let A be a unital associative ring, and let $I \subset A$ be a two-sided nilpotent ideal. Let $\pi : A \rightarrow A/I$ be the quotient map.*

(i) *Every idempotent of A/I lifts to an idempotent of A . That is, if $\bar{e} \in A/I$ satisfies $\bar{e}^2 = \bar{e}$, then there exists an idempotent $e \in A$ such that $\pi(e) = \bar{e}$.*

(ii) *More generally, if*

$$\bar{e} = \bar{e}_1 + \cdots + \bar{e}_n$$

is a decomposition into pairwise orthogonal idempotents in A/I , and $e \in A$ is an idempotent lift of $\bar{e}$, then there exist pairwise orthogonal idempotents $e_1, \dots, e_n \in A$ such that

$$e = e_1 + \cdots + e_n, \quad \pi(e_i) = \bar{e}_i$$

for all $1 \leq i \leq n$.

Theorem 2.33. *Let A be a unital associative ring, and let $I \subset A$ be a two-sided nilpotent ideal. Set $\bar{A} = A/I$, and let $\pi : A \rightarrow \bar{A}$ be the quotient map. If $e \in A$ is an idempotent and $\bar{e} = \pi(e)$, then*

$$e \text{ is primitive in } A \iff \bar{e} \text{ is primitive in } \bar{A}.$$

Lemma 2.34. *Let A be a unital associative ring, and let $f : M \rightarrow N$ be an epimorphism of left A -modules. If M is finitely generated and*

$$\ker f \subset J(A)M,$$

then f is essential. That is, if $L \subset M$ is a submodule such that $f(L) = N$, then $L = M$.

Theorem 2.35. *Let A be a semiperfect ring, let $J = J(A)$, and set $\bar{A} = A/J$.*

(i) *Every finitely generated left A -module M has a projective cover*

$$p : P \twoheadrightarrow M,$$

where P is a finitely generated projective left A -module. Moreover, the induced map

$$P/JP \longrightarrow M/JM$$

is an isomorphism.

(ii) *Let $f : P \twoheadrightarrow M$ be an epimorphism of left A -modules, and suppose that P is finitely generated and projective. Then*

$$f \text{ is essential} \iff \ker f \subset JP.$$

(iii) *If $p : P \twoheadrightarrow M$ is the projective cover in (i), then the induced map*

$$\bar{p} : P/JP \longrightarrow M/JM$$

is a projective cover in $\bar{A}\text{-Mod}$.

3 Week 1

Problem 1

Let R be a ring, and let M be an R -module. Suppose that S is a ring and $\theta : S \rightarrow R$ is a ring homomorphism satisfying $\theta(1_S) = 1_R$. Let S act on M by $sx = \theta(s)x$, where $x \in M, s \in S$. Prove that M is an S -module.

Solution: Define $s \cdot_S x = \theta(s) \cdot_R x$. We verify the axioms:

- $s \cdot_S (x + y) = \theta(s) \cdot_R (x + y) = \theta(s) \cdot_R x + \theta(s) \cdot_R y = s \cdot_S x + s \cdot_S y$.
- $(s_1 + s_2) \cdot_S x = \theta(s_1 + s_2) \cdot_R x = (\theta(s_1) + \theta(s_2)) \cdot_R x = s_1 \cdot_S x + s_2 \cdot_S x$.
- $(s_1 s_2) \cdot_S x = \theta(s_1 s_2) \cdot_R x = \theta(s_1) \cdot_R (\theta(s_2) \cdot_R x) = s_1 \cdot_S (s_2 \cdot_S x)$.
- $1_S \cdot_S x = \theta(1_S) \cdot_R x = 1_R \cdot_R x = x$.

Thus M is an S -module. ■

Problem 2

Let R be a ring, and let M and N be two right R -modules. Let $\text{Hom}_{\mathbb{Z}}(M, N)$ be the set of group homomorphisms from M to N (as abelian groups). The action of R on $\text{Hom}_{\mathbb{Z}}(M, N)$ is given as follows: for any $a \in R, \varphi \in \text{Hom}_{\mathbb{Z}}(M, N)$, define

$$(\varphi a)(x) = \varphi(xa), \quad \forall x \in M.$$

Prove that $\text{Hom}_{\mathbb{Z}}(M, N)$ is a left R -module.

Solution: For any $a, b \in R, \varphi, \psi \in \text{Hom}_{\mathbb{Z}}(M, N)$, and $x \in M$:

- $(a \cdot (\varphi + \psi))(x) = (\varphi + \psi)(x \cdot a) = \varphi(x \cdot a) + \psi(x \cdot a) = (a \cdot \varphi)(x) + (a \cdot \psi)(x)$.
- $((a + b) \cdot \varphi)(x) = \varphi(x \cdot (a + b)) = \varphi(x \cdot a + x \cdot b) = \varphi(x \cdot a) + \varphi(x \cdot b) = (a \cdot \varphi)(x) + (b \cdot \varphi)(x)$.
- $((a \cdot b) \cdot \varphi)(x) = \varphi(x \cdot (a \cdot b)) = \varphi((x \cdot a) \cdot b) = (b \cdot \varphi)(x \cdot a) = (a \cdot (b \cdot \varphi))(x)$.
- $(1_R \cdot \varphi)(x) = \varphi(x \cdot 1_R) = \varphi(x)$, so $1_R \cdot \varphi = \varphi$.

Thus, $\text{Hom}_{\mathbb{Z}}(M, N)$ is a left R -module. ■

Problem 3

Let R be a ring, and let M be an R -module. Denote $\text{Ann}_R M = \{r \in R \mid rm = 0, \forall m \in M\}$. We call M a **faithful** R -module if $\text{Ann}_R M = \{0\}$. Define the action of the quotient ring $R/\text{Ann}_R M$ on M by $\bar{r}x = rx$, where $x \in M, r \in R, \bar{r} = r + \text{Ann}_R M$. Prove that M is a faithful $R/\text{Ann}_R M$ -module.

Solution: Let $I = \text{Ann}_R M$ and $S = R/I$.

1. **M is an S -module:** The action is well-defined because if $\bar{r} = \bar{r}'$, then $r - r' \in I$, so $(r - r') \cdot_R x = 0$, implying $r \cdot_R x = r' \cdot_R x$. The module axioms follow directly from the R -module structure.
2. **M is faithful over S :** Suppose $\bar{r} \in \text{Ann}_S M$. Then for all $x \in M$,

$$\bar{r} \cdot_S x = r \cdot_R x = 0.$$

By definition of I , this implies $r \in I$. Thus $\bar{r} = r + I = \bar{0}$ in S . Since $\text{Ann}_S M = \{\bar{0}\}$, M is a faithful S -module. ■

Problem 4

Let $\varphi : M \rightarrow M$ be a homomorphism of R -modules such that $\varphi\varphi = \varphi$. Prove that

$$M = \ker \varphi \oplus \operatorname{im} \varphi.$$

Solution: Let $\psi = \operatorname{id} - \varphi$. First, we observe that:

$$\varphi \cdot \psi = \varphi \cdot (\operatorname{id} - \varphi) = \varphi - \varphi^2 = \varphi - \varphi = 0,$$

$$\psi \cdot \varphi = (\operatorname{id} - \varphi) \cdot \varphi = \varphi - \varphi^2 = 0.$$

We claim that $M = \operatorname{im} \psi \oplus \operatorname{im} \varphi$ and $\operatorname{im} \psi = \ker \varphi$.

1. For any $x \in M$, we can write:

$$x = \operatorname{id}(x) = (\psi + \varphi)(x) = \psi(x) + \varphi(x).$$

Since $\psi(x) \in \operatorname{im} \psi$ and $\varphi(x) \in \operatorname{im} \varphi$, it is clear that $M = \operatorname{im} \psi + \operatorname{im} \varphi$.

2. Since $\varphi \cdot \psi = 0$, for any $y = \psi(z) \in \operatorname{im} \psi$, we have $\varphi(y) = \varphi(\psi(z)) = 0$. Thus $\operatorname{im} \psi \subseteq \ker \varphi$. If $x \in \ker \varphi$, then $x = \varphi(x) + \psi(x) = \psi(x)$. This implies $x \in \operatorname{im} \psi$, so $\ker \varphi \subseteq \operatorname{im} \psi$. Hence, $\operatorname{im} \psi = \ker \varphi$.

3. Suppose $x \in \operatorname{im} \psi \cap \operatorname{im} \varphi$. Using the identity from step 2:

$$x \in \operatorname{im} \psi \implies x \in \ker \varphi \implies \varphi(x) = 0.$$

Meanwhile, $x \in \operatorname{im} \varphi \implies x = \varphi(y)$ for some $y \in M$. Applying φ to both sides: $\varphi(x) = \varphi^2(y) = \varphi(y) = x$. Since $\varphi(x) = 0$ and $\varphi(x) = x$, we must have $x = 0$.

Therefore, $M = \ker \varphi \oplus \operatorname{im} \varphi$. ■

Problem 5

Show that any R -module is a homomorphic image of some free R -module.

Solution: Let M be an R -module. We construct a free R -module F using the elements of M as a basis. Specifically, let $F = \bigoplus_{m \in M} R \cdot e_m$, where $\{e_m\}_{m \in M}$ is the set of free generators indexed by M .

By the universal property of free modules, there exists a unique R -module homomorphism $\pi : F \rightarrow M$ such that $\pi(e_m) = m$ for all $m \in M$. For any element in F , the map is defined as:

$$\pi \left(\sum_{i=1}^n r_i \cdot e_{m_i} \right) = \sum_{i=1}^n r_i \cdot m_i.$$

For any $m \in M$, there is a corresponding generator $e_m \in F$ such that $\pi(e_m) = m$. This shows that π is a surjective homomorphism. Therefore, $M = \operatorname{im} \pi$ is a homomorphic image of the free R -module F . ■

Problem 6

Let $\mathbb{Q}$ be the field of rational numbers. Is the additive group $(\mathbb{Q}, +)$ a free $\mathbb{Z}$ -module? Explain your answer.

Solution: Suppose $\mathbb{Q}$ is free over $\mathbb{Z}$. For any two non-zero elements $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$ (where $a, b, c, d \in \mathbb{Z} \setminus \{0\}$), they are $\mathbb{Z}$ -linearly dependent since:

$$(b \cdot c) \cdot \frac{a}{b} - (a \cdot d) \cdot \frac{c}{d} = (a \cdot c) - (a \cdot c) = 0.$$

This implies that if $\mathbb{Q}$ were free, its rank must be 1.

However, a free $\mathbb{Z}$ -module of rank 1 is isomorphic to $\mathbb{Z}$ itself, which is a cyclic group. $\mathbb{Q}$ is not cyclic because for any $q \in \mathbb{Q}$, the element $\frac{q}{2}$ cannot be expressed as $n \cdot q$ for any $n \in \mathbb{Z}$. Thus, $\mathbb{Q}$ cannot be a free $\mathbb{Z}$ -module. ■

4 Week 2

Problem 1

Let R be a commutative ring, and suppose that φ is an endomorphism of the free R -module R^m . Prove that if φ is surjective, then φ is injective. Consider that: if φ is injective, is φ surjective?

Solution: Suppose $\{e_1, \dots, e_m\}$ is a basis of R^m and $v_j = \varphi(e_j) = \sum_{i=1}^m a_{ij}e_i$. Then we represent φ by matrix $A = (a_{ij}) \in M_m(R)$, we have

$$(v_1, v_2, \dots, v_m) = (e_1, e_2, \dots, e_m)A. \quad (\clubsuit)$$

Surjectivity implies that $\{e_i\}$ can be linearly represented by $\{v_i\}$. Equivalently, there exists an $m \times m$ matrix $B = (b_{ij}) \in M_m(R)$ such that

$$(e_1, e_2, \dots, e_m) = (v_1, v_2, \dots, v_m)B. \quad (\spadesuit)$$

Combine $\spadesuit$ and $\clubsuit$, we obtain that

$$(e_1, e_2, \dots, e_m) = (e_1, e_2, \dots, e_m)AB.$$

$\{e_i\}$ is a basis implies that $AB = I$ and hence $\det(A)\det(B) = 1$, $\det(A)$ is a unit in R . Let A^* be adjoint matrix of A , then $A^*A = \det(A)I$. Let $c = (c_1, \dots, c_m)^T \in R^m$ such that $\sum c_i v_i = 0$, then

$$(e_1, e_2, \dots, e_m)Ac = (v_1, v_2, \dots, v_m)c = 0$$

implies $Ac = 0$, whence $c = \det(A)^{-1}A^*Ac = 0$. Therefore, $\{v_i\}$, as linearly independent generators of R^m , is a basis. If $w = \sum c_i e_i \in \ker(\varphi)$, then $\sum c_i v_i = 0$ implies $c_i = 0$. The injectivity of φ holds.

The converse is not true. Take $R = \mathbb{Z}$ and consider an injective endomorphism

$$\begin{aligned} \varphi : \mathbb{Z}^m &\rightarrow \mathbb{Z}^m \\ (n_1, \dots, n_m) &\mapsto (2n_1, \dots, 2n_m), \end{aligned}$$

which is not surjective. ■

Problem 2

Let $\varphi : M \rightarrow N$ be a homomorphism of R -modules. Prove that:

1. φ is a monomorphism if and only if for any R -module T and any two R -module homomorphisms $\xi, \eta : T \rightarrow M$, $\varphi\xi = \varphi\eta$ implies $\xi = \eta$;
2. φ is an epimorphism if and only if for any R -module S and any two R -module homomorphisms $\rho, \theta : N \rightarrow S$, $\rho\varphi = \theta\varphi$ implies $\rho = \theta$.

Solution: (1) Let φ be a monomorphism and $\xi, \eta : T \rightarrow M$ be arbitrary module homomorphisms. Let $f = \xi - \eta$, then $\varphi f = 0$ implies $\text{Im}(f) \subseteq \ker \varphi = \{0\}$. Therefore, It follows that $f = 0$ and $\xi = \eta$. Conversely, setting $T = \ker(\varphi)$ and $\xi = \text{id}, \eta = 0$, then $\varphi \xi = \varphi \eta$ implies $\xi = \eta$. Consequently, $\text{id} = 0$ yields $\ker(\varphi) = T = 0$, φ is injective.

(2) Let φ be an epimorphism and $\rho, \theta : N \rightarrow S$ be arbitrary module homomorphisms. Let $g = \rho - \theta$, then $g\varphi = 0$ implies $N = \text{Im}(\varphi) \subseteq \ker(g) \subseteq N$, it follows $\ker(g) = N$ and $g = 0, \rho = \theta$. Conversely, setting $S = \text{coker}(\varphi) = N/\text{Im}(\varphi)$, let ρ be the canonical module homomorphism and θ be a null homomorphism. $\rho\varphi = \theta\varphi$ implies $\rho = \theta$ is a null homomorphism, which yields $\text{Im}(\varphi) = N$. Therefore, the surjectivity of φ holds. ■

Problem 3

Let M and $M_i (1 \leq i \leq n)$ be R -modules. Show that $M \simeq \bigoplus_{i=1}^n M_i$ if and only if for every $i \in \{1, \dots, n\}$, there exists an $\varphi_i \in \text{End}_R(M)$ such that $\text{im } \varphi_i \simeq M_i, \varphi_i \varphi_j = 0$ (for $i \neq j$), and $\varphi_1 + \varphi_2 + \dots + \varphi_n = \text{id}_M$.

Solution: Assume $M \simeq \bigoplus_{i=1}^n M_i$. Without loss of generality, identify M with $\bigoplus_{i=1}^n M_i$. Let $\pi_i : M \rightarrow M_i$ be the natural projection and $\iota_i : M_i \rightarrow M$ be the natural inclusion. Define $\varphi_i = \iota_i \circ \pi_i \in \text{End}_R(M)$. Then $\text{im}(\varphi_i) = \iota_i(M_i) \simeq M_i$ and for $i \neq j$, $\varphi_i \varphi_j = (\iota_i \pi_i)(\iota_j \pi_j) = \iota_i \circ (\pi_i \iota_j) \circ \pi_j = 0$. For any $x = (m_1, \dots, m_n) \in M$, we have $\sum_{i=1}^n \varphi_i(x) = \sum_{i=1}^n (0, \dots, m_i, \dots, 0) = x$. Thus, $\sum_{i=1}^n \varphi_i = \text{id}_M$.

Assume there exist $\varphi_1, \dots, \varphi_n \in \text{End}_R(M)$ satisfying the given conditions. Let $N_i = \text{im } \varphi_i$. By assumption, $N_i \simeq M_i$. Firstly, note that φ_i is idempotent. Multiplying $\sum_{j=1}^n \varphi_j = \text{id}_M$ by φ_i gives $\sum_{j=1}^n \varphi_i \varphi_j = \varphi_i$. Since $\varphi_i \varphi_j = 0$ for $i \neq j$, this reduces to $\varphi_i^2 = \varphi_i$. Consequently, for any $x \in N_i$, $\varphi_i(x) = x$. M is the sum of N_i because for any $x \in M$, $x = \text{id}_M(x) = \sum_{i=1}^n \varphi_i(x)$. Since $\varphi_i(x) \in N_i$, we have $M = \sum_{i=1}^n N_i$. Suppose $\sum_{j=1}^n x_j = 0$ where $x_j \in N_j$. Apply φ_i to both sides:

$$\varphi_i \left(\sum_{j=1}^n x_j \right) = \varphi_i(0) = 0.$$

Because $x_j \in \text{im } \varphi_j$, we can write $x_j = \varphi_j(y_j)$ for some $y_j \in M$. For $j \neq i$, $\varphi_i(x_j) = \varphi_i \varphi_j(y_j) = 0$. Thus, the sum collapses to $\varphi_i(x_i) = 0$. Since $x_i \in N_i$, $x_i = \varphi_i(x_i) = 0$. This holds for all i , proving the sum is direct. Therefore, $M = \bigoplus_{i=1}^n N_i \simeq \bigoplus_{i=1}^n M_i$. ■

Problem 4

1. Prove $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/(m, n)\mathbb{Z}$, where $m, n \in \mathbb{Z}$.
2. Let A be an abelian group, prove $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.
3. Let A, B be two finitely generated abelian groups. What is $A \otimes_{\mathbb{Z}} B$?

Solution: (1) Let $d = (m, n)$. First, consider the map

$$\begin{aligned} f : \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} &\rightarrow \mathbb{Z}/d\mathbb{Z} \\ (a \bmod m, b \bmod n) &\mapsto ab \bmod d. \end{aligned}$$

f is well-defined because if we suppose $a' = a + km$ and $b' = b + ln$ for some integers k, l then $a'b' = (a + km)(b + ln) \equiv ab \bmod d$. Furthermore, multiplication in $\mathbb{Z}/d\mathbb{Z}$ naturally distributes

over addition, making f a $\mathbb{Z}$ -bilinear map. By the universal property of the tensor product, this bilinear map induces a unique $\mathbb{Z}$ -module homomorphism

$$\begin{aligned}\phi : \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} &\rightarrow \mathbb{Z}/d\mathbb{Z} \\ a \otimes b &\mapsto ab \bmod d.\end{aligned}$$

Next, we construct the inverse homomorphism. Consider the $\mathbb{Z}$ -module homomorphism

$$\begin{aligned}\psi : \mathbb{Z}/d\mathbb{Z} &\rightarrow \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \\ c \bmod d &\mapsto c(1 \otimes 1),\end{aligned}$$

which is well-defined because there exist integers x, y such that $xm + yn = d$ by Bézout's identity and thus

$$d(1 \otimes 1) = (xm + yn)(1 \otimes 1) = xm(1 \otimes 1) + yn(1 \otimes 1) = x(m \otimes 1) + y(1 \otimes n) = 0,$$

then $\psi(c + kd) = \psi(c)$. Finally, we verify that ϕ and ψ are mutually inverse. For any $c \bmod d \in \mathbb{Z}/d\mathbb{Z}$, we have $\phi(\psi(c)) = \phi(c(1 \otimes 1)) = c \cdot \phi(1 \otimes 1) = c \cdot 1 = c \bmod d$, so $\phi \circ \psi = \text{id}_{\mathbb{Z}/d\mathbb{Z}}$. Conversely, for any pure tensor $a \otimes b \in \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$, we have $\psi(\phi(a \otimes b)) = \psi(ab) = ab(1 \otimes 1) = a \otimes b$. Since the pure tensors generate the entire tensor product, $\psi \circ \phi = \text{id}_{\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}}$. Therefore, $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/(m, n)\mathbb{Z}$.

(2) First, consider the map

$$\begin{aligned}f : A \times \mathbb{Z}/n\mathbb{Z} &\rightarrow A/nA \\ (a, \bar{k}) &\mapsto ka \bmod nA,\end{aligned}$$

where $\bar{k} = k \bmod n\mathbb{Z}$. To verify this is well-defined, suppose $k' = k + cn$ for some integer c . Then $f(a, \bar{k}') = k'a \bmod nA = ka + cna \bmod nA = ka \bmod nA$, since A is a $\mathbb{Z}$ -module and $cna \in nA$. Furthermore, f naturally preserves addition in both components, making it a $\mathbb{Z}$ -bilinear map. By the universal property of the tensor product, this induces a unique $\mathbb{Z}$ -module homomorphism

$$\begin{aligned}\phi : A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} &\rightarrow A/nA \\ a \otimes \bar{k} &\mapsto ka \bmod nA.\end{aligned}$$

Next, we construct the inverse homomorphism. Consider the module homomorphism

$$\begin{aligned}\psi : A/nA &\rightarrow A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \\ a \bmod nA &\mapsto a \otimes \bar{1},\end{aligned}$$

which is well-defined because if $a' = a + n\alpha$ where $\alpha \in A$ then $\psi(a') = a \otimes \bar{1} + \alpha \otimes \bar{n} = a \otimes \bar{1}$.

Finally, we verify that ϕ and ψ are mutually inverse. For any class $a + nA \in A/nA$, we have $\phi(\psi(a + nA)) = \phi(a \otimes \bar{1}) = 1a \bmod nA = a \bmod nA$, meaning $\phi \circ \psi = \text{id}_{A/nA}$. Conversely, for any pure tensor $a \otimes \bar{k} \in A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$, we can rewrite it using scalar multiplication as $a \otimes \bar{k} = ka \otimes \bar{1}$. Applying the composition yields $\psi(\phi(ka \otimes \bar{1})) = \psi(ka \bmod nA) = ka \otimes \bar{1} = a \otimes \bar{k}$. Since pure tensors generate the entire tensor product, $\psi \circ \phi = \text{id}_{A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}}$. Therefore, $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.

(3) By the Structure Theorem for Finitely Generated Abelian Groups, suppose

$$A \simeq \mathbb{Z}^r \oplus \bigoplus_{i=1}^k \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z}, \quad B \simeq \mathbb{Z}^s \oplus \bigoplus_{j=1}^l \mathbb{Z}/q_j^{\beta_j} \mathbb{Z}$$

where $r, s \geq 0$, $\alpha_i, \beta_j > 0$ and p_i, q_j are prime numbers. Since the tensor product functor distributes over arbitrary direct sums, we can expand $A \otimes_{\mathbb{Z}} B$ into four distinct blocks:

$$A \otimes_{\mathbb{Z}} B \simeq (\mathbb{Z}^r \otimes_{\mathbb{Z}} \mathbb{Z}^s) \oplus \left(\mathbb{Z}^r \otimes_{\mathbb{Z}} \bigoplus_{j=1}^l \mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \right) \oplus \left(\bigoplus_{i=1}^k \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}^s \right) \oplus \left(\bigoplus_{i=1}^k \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \otimes_{\mathbb{Z}} \bigoplus_{j=1}^l \mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \right).$$

By (2) and distributive law, we obtain that

$$\mathbb{Z}^r \otimes_{\mathbb{Z}} \bigoplus_{j=1}^l \mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \simeq \bigoplus_{j=1}^l \left(\mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \right)^r \quad \text{and} \quad \bigoplus_{i=1}^k \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}^s \simeq \bigoplus_{i=1}^k \left(\mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \right)^s.$$

Hence, (1) implies

$$\bigoplus_{i=1}^k \mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \otimes_{\mathbb{Z}} \bigoplus_{j=1}^l \mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \simeq \bigoplus_{p_i=q_j=p} \mathbb{Z}/p^{\min(\alpha_i, \beta_j)} \mathbb{Z}.$$

Therefore we obtain that

$$A \otimes_{\mathbb{Z}} B \simeq \mathbb{Z}^{rs} \oplus \bigoplus_{j=1}^l \left(\mathbb{Z}/q_j^{\beta_j} \mathbb{Z} \right)^r \oplus \bigoplus_{i=1}^k \left(\mathbb{Z}/p_i^{\alpha_i} \mathbb{Z} \right)^s \oplus \bigoplus_{p_i=q_j=p} \mathbb{Z}/p^{\min(\alpha_i, \beta_j)} \mathbb{Z}.$$

Alternatively, if we let $T(A)$ and $T(B)$ denote the torsion subgroups of A and B respectively, this entire result can be represented as

$$A \otimes_{\mathbb{Z}} B \simeq \mathbb{Z}^{rs} \oplus T(B)^r \oplus T(A)^s \oplus (T(A) \otimes_{\mathbb{Z}} T(B)).$$

■

Problem 5

Let $\varphi : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ be the unique monomorphism. Prove that

$$\text{id} \otimes \varphi : \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$$

is a zero homomorphism.

Solution: The unique monomorphism $\varphi : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ must map the generator $\bar{1} \in \mathbb{Z}/2\mathbb{Z}$ to the unique element of order 2 in $\mathbb{Z}/4\mathbb{Z}$, which is $\bar{2}$. Therefore, $\varphi(\bar{1}) = \bar{2}$. Thus, we obtain that

$$(\text{id} \otimes \varphi)(\bar{1} \otimes \bar{1}) = \text{id}(\bar{1}) \otimes \varphi(\bar{1}) = \bar{1} \otimes \bar{2} = 2(\bar{1}) \otimes \bar{1} = 0.$$

Therefore the only non-zero element $\bar{1} \otimes \bar{1}$ in $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ is mapped to 0 meaning that $\text{id} \otimes \varphi$ is a null homomorphism. ■

Problem 6

Let V_1 and V_2 be two vector spaces over a field F with dimension n and m respectively. Let $\{\epsilon_1, \dots, \epsilon_n\}$ and $\{\eta_1, \dots, \eta_m\}$ be bases of V_1 and V_2 respectively. Suppose that $\mathcal{A}$ (resp. $\mathcal{B}$) is an F -linear map of V_1 (resp. V_2), and it is represented by the matrix $A = (a_{ij})_{n \times n}$ (resp. $B = (b_{ij})_{m \times m}$) with respect to the above bases. Prove that

$$\{\epsilon_i \otimes \eta_j \mid 1 \leq i \leq n, 1 \leq j \leq m\}$$

is an F -basis of $V_1 \otimes_F V_2$. Determine the matrix representing the F -linear map $\mathcal{A} \otimes \mathcal{B}$ under this basis.

Solution: We first prove that the set $\{\epsilon_i \otimes \eta_j \mid 1 \leq i \leq n, 1 \leq j \leq m\}$ forms an F -basis for $V_1 \otimes_F V_2$. For arbitrary $v = \sum_{i=1}^n x_i \epsilon_i \in V_1$ and $w = \sum_{j=1}^m y_j \eta_j \in V_2$, we obtain that

$$v \otimes w = \left(\sum_{i=1}^n x_i \epsilon_i \right) \otimes \left(\sum_{j=1}^m y_j \eta_j \right) = \sum_{i=1}^n \sum_{j=1}^m (x_i y_j) (\epsilon_i \otimes \eta_j).$$

Since pure tensors of the form $v \otimes w$ generate the space $V_1 \otimes_F V_2$, and any such pure tensor can be expressed as a linear combination of $\{\epsilon_i \otimes \eta_j\}$, this set spans $V_1 \otimes_F V_2$. To establish linear independence, suppose there exist scalars $c_{ij} \in F$ such that $\sum_{i=1}^n \sum_{j=1}^m c_{ij} (\epsilon_i \otimes \eta_j) = 0$. There are natural isomorphisms

$$\text{Hom}_F(V_1 \otimes_F V_2, F) \simeq \text{Hom}_F(V_1, V_2^*) \simeq V_1^* \otimes_F V_2^*.$$

The first isomorphism arises directly from the Tensor-Hom adjunction and the second holds because of finite dimension of V_1 . Therefore for any $1 \leq k \leq n, 1 \leq l \leq m$, we obtain that

$$c_{kl} = (\epsilon_k^* \otimes \eta_l^*) \left(\sum_{i=1}^n \sum_{j=1}^m c_{ij} (\epsilon_i \otimes \eta_j) \right) = 0$$

meaning that $\{\epsilon_i \otimes \eta_j\}$ is a basis of $V_1 \otimes V_2$.

Next, we determine the matrix representing the F -linear map $\mathcal{A} \otimes \mathcal{B}$ under this basis. By definition, the given matrices $A = (a_{ij})_{n \times n}$ and $B = (b_{ij})_{m \times m}$ describe the action of the linear maps on their respective bases as $\mathcal{A}(\epsilon_j) = \sum_{i=1}^n a_{ij} \epsilon_i$ and $\mathcal{B}(\eta_l) = \sum_{k=1}^m b_{kl} \eta_k$. Evaluating the action of $\mathcal{A} \otimes \mathcal{B}$ on a basis element $\epsilon_j \otimes \eta_l$ yields:

$$(\mathcal{A} \otimes \mathcal{B})(\epsilon_j \otimes \eta_l) = \mathcal{A}(\epsilon_j) \otimes \mathcal{B}(\eta_l) = \left(\sum_{i=1}^n a_{ij} \epsilon_i \right) \otimes \left(\sum_{k=1}^m b_{kl} \eta_k \right) = \sum_{i=1}^n \sum_{k=1}^m (a_{ij} b_{kl}) (\epsilon_i \otimes \eta_k).$$

If we order the nm basis vectors lexicographically (i.e., $(1, 1), (1, 2), \dots, (1, m), (2, 1), \dots$), the resulting $nm \times nm$ matrix representation is the Kronecker product of the matrices

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{pmatrix}$$

■

5 Week 3

Problem 1

Let $M, N, M_i (i \in I)$ and $N_j (j \in J)$ be R -modules. Prove the following isomorphisms of abelian groups.

- (1) $\text{Hom}_R(\bigoplus_{i \in I} M_i, N) \simeq \prod_{i \in I} \text{Hom}_R(M_i, N);$
- (2) $\text{Hom}_R\left(M, \prod_{j \in J} N_j\right) \simeq \prod_{j \in J} \text{Hom}_R(M, N_j).$

Solution: (1) Let $X = \text{Hom}_R(\bigoplus_{i \in I} M_i, N)$ and $Y_i = \text{Hom}_R(M_i, N)$. Let G be an arbitrary abelian group, and suppose we are given a family of group homomorphisms $\varphi_i : G \rightarrow Y_i$ for each $i \in I$. For fixed $g \in G$, the family $\{\varphi_i(g)\}_{i \in I}$ is a collection of R -module homomorphisms from M_i to N . Then by the universal property of the direct sum $\bigoplus_{i \in I} M_i$ in $R\text{-Mod}$, there exists a unique R -module homomorphism f_g such that

$$f_g \circ \iota_i = \varphi_i(g) \quad \text{for all } i \in I.$$

where ι_i are the canonical injections. Define $\Phi : G \rightarrow X$ by setting $\Phi(g) = f_g$. I claim that Φ is the unique group homomorphism such that the diagram

$$\begin{array}{ccc} G & & \\ \varphi_i \downarrow & \searrow \Phi & \\ Y_i & \xleftarrow{\pi_i} & X \end{array}$$

commutes where $\pi_i = \text{Hom}_R(\iota_i, N)$ is the canonical projection induced by contravariant functor $\text{Hom}_R(-, N)$. Let $g, h \in G$. Since $\varphi_i(g + h) = \varphi_i(g) + \varphi_i(h)$ we obtain that

$$f_{g+h} \circ \iota_i = \varphi_i(g) + \varphi_i(h) = (f_g + f_h) \circ \iota_i.$$

Then $f_{g+h} = f_g + f_h$, or equivalently $\Phi(g + h) = \Phi(g) + \Phi(h)$, by the uniqueness condition of the direct sum's universal property. Therefore, Φ is a group homomorphism. To prove the uniqueness of Φ , we suppose there is another group homomorphism $\Psi : G \rightarrow X$ satisfying $(\Psi(g)) \circ \iota_i = \varphi_i(g)$ for all $i \in I$ and all $g \in G$. For any fixed g , both R -module homomorphisms $\Psi(g)$ and $\Phi(g)$ pull back along ι_i to the same maps $\varphi_i(g)$. By the universal property of the direct sum, we must have $\Psi(g) = \Phi(g)$ for all g , so $\Psi = \Phi$. The uniqueness and existence of Φ yields $\text{Hom}_R(\bigoplus_{i \in I} M_i, N) \simeq \prod_{i \in I} \text{Hom}_R(M_i, N)$.

(2) Let $X = \text{Hom}_R(M, \prod_{j \in J} N_j)$ and $Y_j = \text{Hom}_R(M, N_j)$. Let G be an arbitrary abelian group, and suppose we are given a family of group homomorphisms $\varphi_j : G \rightarrow Y_j$ for each $j \in J$. Then for fixed $g \in G$ the family of R -module homomorphisms $\{\varphi_j(g) : M \rightarrow N_j\}_{j \in J}$ uniquely induced an R -module homomorphism $f_g : M \rightarrow \prod_{j \in J} N_j$ such that $\varphi_j(g) = p_j \circ f_g$. Define the map $\Phi : G \rightarrow X$ by setting $\Phi(g) = f_g$. I claim Φ is the unique group homomorphism such that the diagram

$$\begin{array}{ccc} G & & \\ \varphi_j \downarrow & \searrow \Phi & \\ Y_j & \xleftarrow{\pi_j} & X \end{array}$$

commutes where $\pi_j = \text{Hom}_R(M, p_j)$ is the canonical projection induced by covariant functor $\text{Hom}_R(M, -)$. Let $g, h \in G$. Since $\varphi_j(g + h) = \varphi_j(g) + \varphi_j(h)$ we obtain that

$$p_j \circ f_{g+h} = \varphi_j(g) + \varphi_j(h) = p_j \circ (f_g + f_h).$$

It implies $\Phi(g + h) = \Phi(g) + \Phi(h)$ because of the uniqueness condition of the direct product's universal property. Therefore, Φ is a group homomorphism. To prove the uniqueness of Φ , we suppose there is another group homomorphism $\Psi : G \rightarrow X$ satisfying $p_j \circ (\Psi(g)) = \varphi_j(g)$ for all $j \in J$ and all $g \in G$. For any fixed g , both R -module homomorphisms $\Psi(g)$ and $\Phi(g)$ push forward along p_j to the same maps $\varphi_j(g)$. By the universal property of the direct product, we must have $\Psi(g) = \Phi(g)$ for all g , so $\Psi = \Phi$. The uniqueness and existence of Φ yields $\text{Hom}_R(M, \prod_{j \in J} N_j) \simeq \prod_{j \in J} \text{Hom}_R(M, N_j)$. ■

Remark 5.1. The isomorphism established in **(1)** is the direct consequence of the **Hom-Hom adjunction** and the fact that right adjoints preserve limits. Specifically, the contravariant functor $\text{Hom}_R(-, N)$ is the right adjoint of contravariant functor $\text{Hom}_{\mathbf{Ab}}(-, N)$. More explicitly, there is a natural isomorphism

$$\text{Hom}_R(M, \text{Hom}_{\mathbf{Ab}}(A, N)) \cong \text{Hom}_{\mathbf{Ab}}(A, \text{Hom}_R(M, N)),$$

which holds for both left and right R -module categories. Similarly, **(2)** is the direct consequence of the **Tensor-Hom adjunction**. Specifically, the covariant functor $\text{Hom}_R(M, -)$ is the right adjoint of the covariant functor $- \otimes_{\mathbb{Z}} M$ if we consider right R -module category and $M \otimes_{\mathbb{Z}} -$ for the left.

Problem 2

(*Five Lemma*) Suppose that the following diagram of homomorphisms of R -modules

$$\begin{array}{ccccccccc} M_1 & \xrightarrow{f_1} & M_2 & \xrightarrow{f_2} & M_3 & \xrightarrow{f_3} & M_4 & \xrightarrow{f_4} & M_5 \\ \downarrow \varphi_1 & & \downarrow \varphi_2 & & \downarrow \varphi_3 & & \downarrow \varphi_4 & & \downarrow \varphi_5 \\ N_1 & \xrightarrow{g_1} & N_2 & \xrightarrow{g_2} & N_3 & \xrightarrow{g_3} & N_4 & \xrightarrow{g_4} & N_5 \end{array}$$

is commutative and has exact rows. Prove that

- (1) if φ_1 is surjective, and φ_2 and φ_4 are injective, then φ_3 is injective;
- (2) if φ_5 is injective, and φ_2 and φ_4 are surjective, then φ_3 is surjective.

Solution: **(1)** Suppose $a_3 \in M_3$ such that $\varphi_3(a_3) = 0$, then $\varphi_4 \circ f_3(a_3) = g_3 \circ \varphi_3(a_3) = 0$ implies $f_3(a_3) = 0$ since φ_4 is injective. Therefore, $\text{Im}(f_2) = \ker(f_3)$ implies that there exists an element $a_2 \in M_2$ such that $f_2(a_2) = a_3$. Let $b_2 = \varphi_2(a_2)$, then $g_2(b_2) = \varphi_3 \circ f_2(a_2) = \varphi_3(a_3) = 0$ implies there exists $b_1 \in N_1$ such that $g_1(b_1) = b_2$. Because φ_1 is surjective then there exists a_1 such that $\varphi_1(a_1) = b_1$. I claim that $a_2 = f_1(a_1)$ and therefore $a_2 \in \text{Im}(f_1) = \ker(f_2)$ implies $a_3 = f_2(a_2) = 0$ and thus φ_3 is injective. To prove it, we can consider $\varphi_2(a_2 - f_1(a_1)) = \varphi_2(a_2) - g_1 \circ \varphi_1(a_1) = b_2 - b_2 = 0$ which implies $a_2 - f_1(a_1) = 0$ because of injectivity of φ_2 .

(2) Suppose b_3 be arbitrary elements in N_3 respectively. Let $b_4 = g_3(b_3)$. Because φ_4 is surjective then we can let $a_4 \in M_4$ such that $\varphi_4(a_4) = b_4$. Then $\varphi_5 \circ f_4(a_4) = g_4 \circ \varphi_4(a_4) = g_4 \circ g_3(b_3) = 0$ implies $f_4(a_4) = 0$ because of injectivity of φ_5 . Then we can suppose $a_3 \in M_3$ such that $f_3(a_3) = a_4$. Then $g_3(b_3 - \varphi_3(a_3)) = b_4 - \varphi_4(a_4) = 0$ implies there exists $b_2 \in N_2$ such that $g_2(b_2) = b_3 - \varphi_3(a_3)$. Then the surjectivity of φ_2 implies there exists a_2 such that $\varphi_2(a_2) = b_2$. Since diagram commutes, we obtain that $\varphi_3 \circ f_2(a_2) = b_3 - \varphi_3(a_3)$, or equivalently, $b_3 = \varphi_3(f_2(a_2) + a_3) \in \text{Im}(\varphi_3)$ which implies φ_3 is surjective. ■

Problem 3

(*Snake Lemma*) Suppose that the following diagram of homomorphisms of R -modules

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 \\ & & \downarrow \varphi' & & \downarrow \varphi & & \downarrow \varphi'' & & \\ 0 & \longrightarrow & N' & \xrightarrow{\mu} & N & \xrightarrow{\nu} & N'' & \longrightarrow & 0 \end{array}$$

is commutative and has exact rows. Prove that

$$0 \rightarrow \ker \varphi' \xrightarrow{\alpha'} \ker \varphi \xrightarrow{\beta'} \ker \varphi'' \xrightarrow{\delta} \operatorname{coker} \varphi' \xrightarrow{\mu'} \operatorname{coker} \varphi \xrightarrow{\nu'} \operatorname{coker} \varphi'' \rightarrow 0$$

is a long exact sequence, where $\alpha', \beta', \mu', \nu'$ are homomorphisms induced by α, β, μ, ν , respectively. (Hint: for any $x \in \ker \varphi''$, take $y \in M$ such that $\beta(y) = x$, and take $z \in N'$ such that $\mu(z) = \varphi(y)$. Define $\delta(x) = \bar{z} = z + \operatorname{im} \varphi'$)

Solution: α' is well-defined and injective. Suppose $m' \in \ker(\varphi')$, since diagram commutes, we obtain that $\varphi \circ \alpha(m') = \mu \circ \varphi'(m') = 0$ which implies $\alpha(m') \in \ker(\varphi)$. Therefore, $\alpha' = \alpha|_{\ker(\varphi')} : \ker(\varphi') \rightarrow \ker(\varphi)$ is well-defined. Hence, α' is injective since α is injective. β' is well-defined and $\operatorname{Im}(\alpha') = \ker(\beta')$. Let $m \in \ker(\varphi)$, then $\varphi'' \circ \beta(m) = \nu \circ \varphi(m) = 0$ yields well-definedness. Suppose $m' \in \ker(\varphi')$ then $\beta' \circ \alpha'(m') = \beta \circ \alpha(m') = 0$ where $\beta \circ \alpha = 0$ because of the exactness of the first row. Therefore, we obtain that $\operatorname{Im}(\alpha') \subseteq \ker(\beta')$. Conversely, if $m \in \ker(\beta')$, then $\beta(m) = \beta'(m) = 0$. By exactness, there exists m' such that $\alpha(m') = m$. Hence, $m \in \ker(\varphi)$ implies that $\mu \circ \varphi'(m') = \varphi \circ \alpha(m') = \varphi(m) = 0$. Because μ is injective, then we obtain that $m' \in \ker(\varphi')$. Therefore, for an arbitrary element $m \in \ker(\beta')$ there exists $m' \in \ker(\varphi')$ such that $m = \alpha'(m')$ meaning that $\ker(\beta') \subseteq \operatorname{Im}(\alpha')$. δ is well-defined. Because β is surjective then for any $x \in \ker(\varphi'')$ there exists an element $y \in M$ such that $\beta(y) = x$. Since $x \in \ker(\varphi'')$, we obtain that $\nu \circ \varphi(y) = \varphi''(x) = 0$, which implies $\varphi(y) \in \ker(\nu)$. By exactness, there exists an element $z \in N'$ such that $\mu(z) = \varphi(y)$. Let $y_0, y_1 \in M$ be arbitrary two elements such that $\beta(y_0) = \beta(y_1) = x$. Let $z_0, z_1 \in N'$ be arbitrary two elements such that $\mu(z_0) = \varphi(y_0)$ and $\mu(z_1) = \varphi(y_1)$. Because $\beta(y_1 - y_0) = 0$, then there exists an element $m' \in M'$ such that $\alpha(m') = y_1 - y_0$. Therefore,

$$\begin{aligned} \mu(\varphi'(m') - z_1 + z_0) &= \mu \circ \varphi'(m') - \mu(z_1 - z_0) \\ &= \varphi \circ \alpha(m') - \mu(z_1 - z_0) \\ &= \varphi(y_1 - y_0) - \mu(z_1 - z_0) = 0 \end{aligned}$$

implies $\varphi'(m') = z_1 - z_0$ since μ is injective. Equivalently, we obtain that $\overline{z_1 - z_0} = 0$ which yields the well-definedness of δ . $\operatorname{Im}(\beta') = \ker(\delta)$. Suppose $m \in \ker(\varphi)$, then we can select $z = 0 \in N'$ because $\mu(z) = \varphi(m) = 0$ which implies $\delta \circ \beta'(m) = 0$. Therefore, $\operatorname{Im}(\beta') \subseteq \ker(\delta)$. Suppose $m'' \in \ker(\delta)$, then there exists $m \in M$ and $n' \in N'$ such that $\beta(m) = m''$ and $\varphi(m) = \mu(n')$. Hence $\delta(m'') = 0$ implies there exists an element $m' \in M'$ such that $\varphi'(m') = n'$. I claim that $m - \alpha(m') \in \ker(\varphi)$ and thus $\beta'(m - \alpha(m')) = \beta(m) - \beta \circ \alpha(m') = m''$ implies that $\ker(\delta) \subseteq \operatorname{Im}(\beta')$. To see that, we consider $\varphi(m - \alpha(m')) = \varphi(m) - \mu \circ \varphi'(m') = \varphi(m) - \mu(n') = 0$. μ' is well-defined and $\operatorname{Im}(\delta) = \ker(\mu')$. Since the diagram commutes, $\mu \circ \varphi' = \varphi \circ \alpha$ implies $\mu(\operatorname{Im}(\varphi')) \subseteq \operatorname{Im}(\varphi)$ which yields well-definedness. Let m'' be an arbitrary element in $\ker(\varphi'')$ then there exists $m \in M$ and $n' \in N'$ such that $\beta(m) = m''$ and $\varphi(m) = \mu(n')$. Then we have $\mu' \circ \delta(m'') = \mu'(n' + \operatorname{Im}(\varphi')) = \mu(n') + \operatorname{Im}(\varphi) = 0 \in \operatorname{coker}(\varphi)$, which implies $\operatorname{Im}(\delta) \subseteq \ker(\mu')$. For an arbitrary element $\bar{n}' \in \ker(\mu')$, then we have $\mu(n') \in \operatorname{Im}(\varphi)$, which implies there exists an element $m \in M$ such that $\varphi(m) = \mu(n')$. Let $m'' = \beta(m)$ and I claim that $m'' \in \ker(\varphi'')$ and thus $\delta(m'') = \bar{n}'$ implies $\ker(\mu') \subseteq \operatorname{Im}(\delta)$. To see that, just consider $\varphi''(m'') = \nu \circ \varphi(m) = \nu \circ \mu(n') = 0$. ν' is well-defined and $\ker(\nu') = \operatorname{Im}(\mu')$. Similarly, because diagram commutes, then $\nu \circ \varphi = \varphi'' \circ \beta$ implies that $\nu(\operatorname{Im}(\varphi)) \subseteq \operatorname{Im}(\varphi'')$ meaning that ν' is well-defined. Let $\bar{n}'$ be an arbitrary element in $\operatorname{coker}(\varphi')$, then $\nu' \circ \mu'(\bar{n}') = \nu'(\mu(n') + \operatorname{Im}(\varphi)) = \nu \circ \mu(n') + \operatorname{Im}(\varphi'') = 0 \in \operatorname{coker}(\varphi'')$. Therefore, $\operatorname{Im}(\mu') \subseteq \ker(\nu')$. Let $\bar{n}$ be an arbitrary element such that $\nu'(\bar{n}) = 0$, equivalently $\nu(n) \in \operatorname{Im}(\varphi'')$. Then there exists an element $m'' \in M''$ such that $\varphi''(m'') = \nu(n)$. Since β is surjective, then there exists an element $m \in M$ such that $\beta(m) = m''$. I claim that $n - \varphi(m) \in \ker(\nu)$ and thus there exists an element $n' \in N'$ such that

$\mu(n') = n - \varphi(m)$, which implies that $\mu'(\bar{n'}) = \bar{n}$ meaning that $\ker(\nu') \subseteq \text{Im}(\mu')$. To see that, we consider $\nu(n - \varphi(m)) = \nu(n) - \varphi'' \circ \beta(m) = \nu(n) - \varphi''(m) = 0$. **ν' is surjective.** It is directly because ν' is well-defined and induced by surjection ν . Explicitly, for any arbitrary element $\bar{n''} \in \text{coker}(\varphi'')$, there exists an element $n \in N$ such that $\nu(n) = n''$. Since $\nu(\text{Im}(\varphi)) \subseteq \text{Im}(\varphi'')$ we have discussed, then we have $\nu'(\bar{n}) = n'' + \text{Im}(\varphi'') = \bar{n''}$. ■

Lemmas for Problem 3

Lemma 5.2. *A Let $f : M \rightarrow M''$ and $g : K \rightarrow M''$ be R -module homomorphisms. Let $P = M \times_{M''} K$ be the pullback of f along g , defined explicitly by the subset of the direct sum:*

$$P = \{(m, k) \in M \times K \mid f(m) = g(k)\}$$

equipped with the canonical projections $\pi_M : P \rightarrow M, (m, k) \mapsto m$ and $\pi_K : P \rightarrow K, (m, k) \mapsto k$. Then the following properties hold:

1. *If f is surjective, then the projection π_K is surjective.*
2. *If f is injective, then the projection π_K is injective.*

Proof. (1) Suppose f is surjective. Let $k \in K$ be an arbitrary element. Because f is surjective, then there exists an element $m \in M$ such that $f(m) = g(k)$, which implies $(m, k) \in P$ then $k = \pi_K(m, k)$ implies surjectivity.

(2) Suppose f is injective. Suppose $(m, k) \in P$ such that $\pi_K(m, k) = 0$. Then we obtain that $f \circ \pi_M(m, k) = f(m) = g(k) = g \circ \pi_K(m, k) = 0$. The injectivity of f implies $m = 0$. Hence, $k = \pi_K(m, k) = 0$ implies the injectivity of π_K . □

Lemma 5.3. *B Let $\mu : N' \rightarrow N$ and $h : N' \rightarrow L$ be R -module homomorphisms. Let $Q = N \oplus_{N'} L$ be the pushout of μ along h , defined explicitly as the quotient module of the direct sum:*

$$Q = (N \oplus L)/W$$

where $W = \{(\mu(n'), -h(n')) \mid n' \in N'\}$. The induced maps are given by $\bar{h} : N \rightarrow Q, n \mapsto \overline{(n, 0)}$ and $i_L : L \rightarrow Q, l \mapsto \overline{(0, l)}$. Then the following properties hold:

1. *If μ is surjective, then the induced map i_L is surjective.*
2. *If μ is injective, then the induced map i_L is injective.*

Proof. (1) Suppose μ is surjective. It suffices to prove that any pair with form $\overline{(n, 0)}$ has preimage because $\overline{(n, l)} = \overline{(n, 0)} + \overline{(0, l)}$ and $\overline{(0, l)}$ has preimage. Because μ is surjective, then there exists $n' \in N'$ such that $\mu(n') = n$. I claim that $\overline{(n, 0)} = \overline{(0, h(n'))} = i_L(h(n'))$ and thus surjectivity of i_L holds. To see that, we consider $\overline{(n, 0)} - \overline{(0, h(n'))} = \overline{(n, -h(n'))} = 0$ since $n = \mu(n')$.

(2) Suppose μ is injective. Let $l \in L$ be an arbitrary element such that $i_L(l) = \overline{(0, l)} = 0$, or equivalently there exists $n' \in N'$ such that $\mu(n') = 0$ and $-h(n') = l$. The injectivity of μ implies $n' = 0$ and hence $l = -h(n') = 0$. Therefore, i_L is injective. □

6 Week 4

Lemmas for Problem

Lemma 6.1. *A Let $\mu : N' \rightarrow N$ and $h : N' \rightarrow L$ be R -module homomorphisms. Let $Q = N \oplus_{N'} L$ be the pushout of μ along h , defined explicitly as the quotient module of the direct sum:*

$$Q = (N \oplus L)/W$$

where $W = \{(\mu(n'), -h(n')) \mid n' \in N'\}$. The induced maps are given by $\bar{h} : N \rightarrow Q, n \mapsto \overline{(n, 0)}$ and $i_L : L \rightarrow Q, l \mapsto \overline{(0, l)}$. Then the following properties hold:

1. If μ is surjective, then the induced map i_L is surjective.
2. If μ is injective, then the induced map i_L is injective.

Proof. (1) Suppose μ is surjective. It suffices to prove that any pair with form $\overline{(n, 0)}$ has preimage because $\overline{(n, l)} = \overline{(n, 0)} + \overline{(0, l)}$ and $\overline{(0, l)}$ has preimage. Because μ is surjective, then there exists $n' \in N'$ such that $\mu(n') = n$. I claim that $\overline{(n, 0)} = \overline{(0, h(n'))} = i_L(h(n'))$ and thus surjectivity of i_L holds. To see that, we consider $\overline{(n, 0)} - \overline{(0, h(n'))} = \overline{(n, -h(n'))} = 0$ since $n = \mu(n')$.

(2) Suppose μ is injective. Let $l \in L$ be an arbitrary element such that $i_L(l) = \overline{(0, l)} = 0$, or equivalently there exists $n' \in N'$ such that $\mu(n') = 0$ and $-h(n') = l$. The injectivity of μ implies $n' = 0$ and hence $l = -h(n') = 0$. Therefore, i_L is injective. $\square$

Lemma 6.2. *Let $f : M \rightarrow M''$ and $g : K \rightarrow M''$ be R -module homomorphisms. Let $P = M \times_{M''} K$ be the pullback of f along g , defined explicitly by the subset of the direct sum:*

$$P = \{(m, k) \in M \times K \mid f(m) = g(k)\}$$

equipped with the canonical projections $\pi_M : P \rightarrow M, (m, k) \mapsto m$ and $\pi_K : P \rightarrow K, (m, k) \mapsto k$. Then the following properties hold:

1. If f is surjective, then the projection π_K is surjective.
2. If f is injective, then the projection π_K is injective.

Proof. (1) Suppose f is surjective. Let $k \in K$ be an arbitrary element. Because f is surjective, then there exists an element $m \in M$ such that $f(m) = g(k)$, which implies $(m, k) \in P$ then $k = \pi_K(m, k)$ implies surjectivity.

(2) Suppose f is injective. Suppose $(m, k) \in P$ such that $\pi_K(m, k) = 0$. Then we obtain that $f \circ \pi_M(m, k) = f(m) = g(k) = g \circ \pi_K(m, k) = 0$. The injectivity of f implies $m = 0$. Hence, $k = \pi_K(m, k) = 0$ implies the injectivity of π_K . $\square$

Problem 1

Let J be an R -module. Show that the following are equivalent.

- (1) J is an injective module.
- (2) For any R -module monomorphism $\varphi : J \rightarrow M$, there exists an R -module homomorphism $\psi : M \rightarrow J$ such that $\psi\varphi = \text{id}_J$.
- (3) If J is a submodule of some R -module M , then J is a direct summand of M .

Solution: We first prove that (1) implies (2). According to the definition of injective module, for morphism id_J and monomorphism $\varphi : J \rightarrow M$, there exists $\psi : M \rightarrow J$ such that diagram

$$\begin{array}{ccc} J & \xhookrightarrow{\varphi} & M \\ & \searrow \text{id}_J & \downarrow \exists \psi \\ & & J \end{array}$$

commutes. In other words, $\psi \circ \varphi = \text{id}_J$.

Next, we prove that (2) implies (3). Because J is the submodule of M , then we obtain the canonical injection $\varphi : J \hookrightarrow M$, therefore (2) implies that there exists a morphism $\psi : M \rightarrow J$ such that $\psi \circ \varphi = \text{id}_J$. I claim that $M \cong J \oplus \ker(\psi)$. To see that, we define maps

$$\begin{aligned} f : J \oplus \ker(\psi) &\rightarrow M \\ (j, h) &\mapsto \varphi(j) + h \end{aligned}$$

and

$$\begin{aligned} g : M &\rightarrow J \oplus \ker(\psi) \\ m &\mapsto (\psi(m), m - \varphi \circ \psi(m)). \end{aligned}$$

f is a R -module morphism is quite clear because we have

$$\text{Hom}_R(J \oplus \ker(\varphi), M) \cong \text{Hom}_R(J, M) \oplus \text{Hom}_R(\ker(\varphi), M).$$

We need to verify g is a R -module morphism. The well-definedness of g is given by the fact that

$$\psi(m - \varphi \circ \psi(m)) = \psi(m) - \psi \circ \varphi \circ \psi(m) = 0,$$

and g is a R -module homomorphism since ψ and φ are R -module homomorphism. I claim that f, g are bijection and $f^{-1} = g$. To see that, we just need to consider the composition of two maps

$$f \circ g(m) = \varphi \circ \psi(m) + (m - \varphi \circ \psi(m)) = m$$

and

$$\begin{aligned} g \circ f(j, h) &= (\psi(\varphi(j) + h), \varphi(j) + h - \varphi \circ \psi(\varphi(j) + h)) \\ &= (j + \psi(h), \varphi(j) + h - \varphi(j) + \varphi \circ \psi(h)) \\ &= (j, h). \end{aligned}$$

Therefore, $M \cong J \oplus \ker(\psi)$ and then J is a direct summand of M .

Now we prove (3) implies (1). Suppose $\varphi : M \rightarrow N$ is an arbitrary monomorphism and $f : M \rightarrow J$ is an arbitrary homomorphism. Use the concept already defined in **Lemma A**. We consider the pushout f along φ

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & N \\ f \downarrow & & \downarrow j \\ J & \xrightarrow{i} & J \oplus_M N \end{array}$$

Since φ is injective, i is injective as well. Therefore, J can be interpreted as the submodule of $J \oplus_M N$ by canonical inclusion. Hence, (3) implies that J is the direct summand of $J \oplus_M N$, then we can assume that $J \oplus_M N \cong J \oplus K$. Define map $\psi = p \circ j : N \rightarrow J$ where $p : J \oplus K \rightarrow J$ is the canonical projection. Then $f = \psi \circ \varphi$ since $f = p \circ i \circ f = p \circ j \circ \varphi = \psi \circ \varphi$. ■

Problem 2

Suppose that M_i ($i \in I$) are R -modules. Prove that $\bigoplus_{i \in I} M_i$ is a flat R -module if and only if each M_i , $i \in I$ is a flat R -module.

Solution: Let $M = \bigoplus_{i \in I} M_i$. Let $f : A \hookrightarrow B$ be an arbitrary monomorphism. Assume that M_i are all flat, then by definition, the R -module morphisms $f \otimes_R \text{id}_{M_i} : A \otimes_R M_i \hookrightarrow B \otimes_R M_i$ are all monomorphic. Therefore

$$\bigoplus_{i \in I} (f \otimes_R \text{id}_{M_i}) : \bigoplus_{i \in I} (A \otimes_R M_i) \hookrightarrow \bigoplus_{i \in I} (B \otimes_R M_i)$$

is monomorphism by definition of direct sum. Hence, the commutativity of tensor product and direct sum implies that

$$f \otimes_R \text{id}_M : A \otimes_R M \hookrightarrow B \otimes_R M$$

is monomorphism meaning that M is flat.

Conversely, we use concepts defined in **Lemma B**. For arbitrary index i , consider canonical injection $j : B \otimes_R M_i \rightarrow B \otimes_R M$ pullback along $f \otimes_R \text{id}_M$. Let $P = (B \otimes_R M_i) \times_{B \otimes_R M} (A \otimes_R M)$, and according **Lemma B**, the pullback homomorphism $g : P \rightarrow B \otimes_R M_i$ is monomorphism. Let $\iota : A \otimes_R M_i \rightarrow P$ by sending pure tensor $a \otimes m_i$ to $(f(a) \otimes m_i, a \otimes m_i)$ and extending it linearly. It is clear that ι is well-defined because $f \otimes_R \text{id}_M(a \otimes m_i) = f(a) \otimes m_i$ and for arbitrary $r \in R$, we have

$$r \cdot \iota(a \otimes m_i) = (f(r \cdot a) \otimes m_i, r \cdot a \otimes m_i) = \iota(r \cdot a \otimes m_i)$$

and other requirements of R -module is easy to prove. Therefore ι is a R -module morphism, hence it is injective since $a \otimes m_i$ equals zero if and only if $(f(a) \otimes m_i, a \otimes m_i)$ equals zero. Therefore $f \otimes_R \text{id}_{M_i} = g \circ \iota$ is monomorphic meaning that M_i is flat. ■

Problem 3

Prove: the following three conditions on an R -module V are equivalent.

- (1) (Ascending chain condition.) For every infinite chain of R -submodules of V ,

$$V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n \subseteq \cdots,$$

there is a number m such that $V_m = V_{m+1} = \cdots$.

- (2) Any non-empty set of submodules of V contains a maximal element with respect to inclusion.
- (3) Every R -submodule of V is finitely generated.

Solution: First we show that (1) implies (2). If there exists a non-empty set of submodules of V , denoted by $\mathcal{S}$, such that no maximal element exists. Then for every submodule $V_1 \in \mathcal{S}$, there exists a submodule $V_2 \in \mathcal{S}$ such that $V_1 \subsetneq V_2$. Continuing this process, we obtain an infinite ascending chain contradicting to (1).

Now we prove (2) implies (3). Let W be an arbitrary submodule of V . Let $\mathcal{S}$ be the collection of all finitely generated submodules of W . By (2), $\mathcal{S}$ contains a maximal element W' . Suppose $W' \subsetneq W$ is a proper submodule, then select an element $\alpha \in W \setminus W'$ and let W'' be the submodule generated by W' and α , which is finitely generated because W' is finitely generated. However, W' is the maximal element of $\mathcal{S}$ meaning that $W'' \notin \mathcal{S}$, which is a contradiction. Therefore, $W' = W$ implies that W is finitely generated and thus every submodule of V is finitely generated.

Finally, we claim that (3) implies (1). If there exists an infinite ascending chain

$$V_1 \subsetneq V_2 \subsetneq \cdots \subsetneq V_n \subsetneq \cdots,$$

By (3), the submodule $W = \cup_{i=1}^{\infty} V_i$ must be finitely generated. For arbitrary two elements $x, y \in W$, there exists an integer n such that $x, y \in V_1 \cup \cdots \cup V_n = V_n$ by definition. Then all requirements of R -module are satisfied since V_n is a submodule. Suppose $W = \langle w_1, w_2, \dots, w_m \rangle$ is generated by finite elements w_i , then there exists an integer s such that $w_i \in V_1 \cup \cdots \cup V_s = V_s$ for all i meaning that $W \subseteq V_s$ and therefore for every $k \geq s$ we obtain that $V_s = V_k$, which is a contradiction. ■

Problem 4

Let W be an R -submodule of V . Prove: V is Noetherian if and only if both W and V/W are Noetherian.

Solution: Suppose V is Noetherian. Every ascending chain of W is also the ascending chain of V meaning that chain is stationary. Hence, Correspondence Theorem implies that if

$$\overline{V}_1 \subseteq \overline{V}_2 \subseteq \cdots \quad (4)$$

is an ascending chain of V/W , there exists ascending chain of V

$$V_1 \subseteq V_2 \subseteq \cdots \quad (\textcircled{2})$$

such that V_i is the lift of $\overline{V}_i$. The chain $(\textcircled{2})$ is stationary implies chain (4) is stationary because $\overline{V}_i = V_i + W$.

Conversely, suppose W and V/W are Noetherian and let

$$V_1 \subseteq V_2 \subseteq \cdots \quad (\text{II})$$

be an arbitrary ascending chain of V . Since V/W is Noetherian, the ascending chain

$$V_1 + W \subseteq V_2 + W \subseteq \cdots \quad (\text{III})$$

is stationary and we suppose it stationary at n . If chain (II) is not stationary, then select elements $\alpha_i \in V_i \setminus V_{i-1}$ for $i > n$. Chain (III) stationary implies that $\alpha_i \in V_i + W = V_n + W$. Then we can suppose $\alpha_i = \beta_i + \gamma_i$ where $\beta_i \in V_n$ and $\gamma_i \in W$. Hence, $V_n \subseteq V_i$ implies that $\gamma_i \in V_i$ and $\gamma_i \notin V_{i-1}$. If not, α_i must be the element contained in V_{i-1} contradicting our assumption. Consider ascending chain of W

$$\langle \gamma_{n+1} \rangle \subseteq \langle \gamma_{n+1}, \gamma_{n+2} \rangle \subseteq \cdots$$

which is stationary since W is Noetherian. Suppose it is stationary at m meaning that γ_{m+1} can be linearly represented by $\gamma_{n+1}, \dots, \gamma_m$ and therefore $\gamma_{m+1} \in V_m$, which is a contradiction, chain (II) must be stationary. ■

Problem 5

Let V be a Noetherian R -module and $f : V \rightarrow V$ an epimorphism. Show that f is an R -isomorphism.

Solution: Let $W_i = \ker(f^i)$ and consider an ascending chain

$$W_1 \subseteq W_2 \subseteq \dots \quad (\text{II})$$

because V is Noetherian then this chain is stationary. Suppose it is stationary at n . Then W_n is the submodule of V consisting exactly of all nilpotent elements associated to f . I claim that $f(W_n) \subseteq W_n$ and for every element $x \in V \setminus W_n$ we have $f(x) \notin W_n$. To see that, suppose $w \in W_n$ is nilpotent, then $f(w)$ is nilpotent as well, meaning that $f(w) \in W_n$. Hence, for arbitrary element $x \in V \setminus W_n$, if $f(x) \in W_n$ then it implies that $f(x)$ is nilpotent and thus x is nilpotent, which is contradiction. Suppose that $n > 1$, ascending chain (II) stationary at n means that $W_{n-1} \subsetneq W_n$, then $W_n \setminus W_{n-1}$ has no preimage of f because $f(W_n) = W_{n-1}$. It is contradicts the assumption that f is epimorphism. If $n = 1$ and $W_1 \neq \{0\}$ then $W_1 \setminus \{0\}$ has no preimage. Suppose $x \in V$ such that $f(x) \in W_1 \setminus \{0\}$, then $f^2(x) = 0$ implies that $x \in W_2$. However, $W_2 = W_1$ implies that $f(x) = 0$, which is a contradiction. Therefore, f must be a monomorphism, hence an isomorphism. ■

7 Week 5

Problem 1

Are the following statements correct?

- (1) Any submodule of a semisimple module is also semisimple.
- (2) Any semisimple Noetherian module is Artinian.

Solution: Both statements are correct. (1) Suppose N is an arbitrary submodule of semisimple module M . Then by discussion in our class, there exists simple submodules $\{M_i\}_{i \in I}$ such that

$$M = \bigoplus_{i \in I} M_i$$

and subindex set J of I such that

$$M = N \oplus \left(\bigoplus_{j \in J} M_j \right)$$

Consider the canonical projection $M \rightarrow N$ then by isomorphism theorem we obtain that

$$N \cong M / \left(\bigoplus_{j \in J} M_j \right) \cong \bigoplus_{i \in I \setminus J} M_i$$

meaning that N is semisimple.

- (2) Suppose M is an arbitrary semisimple Noetherian module with direct sum decomposition

$$M = \bigoplus_{i \in I} M_i.$$

Noetherian implies that M is finite-generated. Without lose of generality, we can suppose that $x_1, \dots, x_n$ are generators and $x_j \in M_j$. Then M admits a finite decomposition

$$M = \bigoplus_{j \in J} M_j.$$

where $J = \{1, \dots, n\}$ is finite. Therefore M has a finite composition series meaning that M is Artinian. ■

Problem 2

Prove that: the following conditions on an R -module V with composition series are equivalent.

- (1) V is semisimple.
- (2) Every irreducible submodule of V is a direct summand of V .
- (3) Every maximal submodule of V is a direct summand of V .

Solution: (1) implies (2) and (3) just by equivalent definition of semisimple module we have discussed in our class.

Next, we prove that (2) implies (1). Suppose $V \neq \langle 0 \rangle$, otherwise it is trivially semisimple. Since V has a composition series, it must contain an irreducible submodule. let us denote it by V_1 . Then by (2), we can let $V = V_1 \oplus V_1'$. We claim that for every irreducible submodule of V_1' also is a direct summand of V_1' . To see that, we suppose V_2 is an irreducible submodule of V_1' and hence it is also an irreducible submodule of V , then we can suppose that $V = V_2 \oplus V_2'$. Then we claim that

$$V_1' = V_2 \oplus (V_2' \cap V_1').$$

It is obvious that $V_2 \cap (V_2' \cap V_1') = 0$ and $V_2 \oplus (V_2' \cap V_1') \subseteq V_1'$. For $x \in V_1'$, we can suppose that $x = y + z$ where $y \in V_2, z \in V_2'$. Since $x, y \in V_1'$ then $z = x - y \in V_1'$ which implies that $V_1' \subseteq V_2 \oplus (V_2' \cap V_1')$. Continuing this process we obtain a decomposition

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_k \oplus W_k$$

where $V_1, \dots, V_k$ are irreducible. Suppose that the length of composition series of V is n , then we obtain that $W_n = 0$, because in this case we have a strict ascending chain

$$\langle 0 \rangle \subsetneq V_1 \subsetneq \cdots \subsetneq V_1 \oplus V_2 \oplus \cdots \oplus V_n$$

which implies that

$$V = \bigoplus_{i=1}^n V_i$$

and $W_n = \langle 0 \rangle$. Then V is semisimple.

We now show that (3) implies (1). Similarly, we suppose that V_1' is an arbitrary maximal submodule of V . Then we suppose that $V = V_1 \oplus V_1'$ and $V_1 \cong V/V_1'$ implies that V_1 is irreducible. We claim that every maximal submodule of V_1' , denoted by V_2' , is a direct summand of V_1' . To see that, we note that $V_1 \oplus V_2'$ is a maximal submodule of V since

$$V/(V_1 \oplus V_2') \cong (V_1 \oplus V_1')/(V_1 \oplus V_2') \cong V_1'/V_2'.$$

Let $V_1' = V_2 \oplus V_2'$ and continuing this process, then we obtain a decomposition

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_n \oplus W_n$$

and by same reason we have discussed $W_n = \langle 0 \rangle$ and V is semisimple. ■

Problem 3

Prove: $\text{Soc}(U \oplus V) = \text{Soc}(U) \oplus \text{Soc}(V)$.

Solution: It is obvious that

$$\text{Soc}(U \oplus V) \supseteq \text{Soc}(U) \oplus \text{Soc}(V).$$

Because if M is an irreducible submodule of U or V , it is also an irreducible submodule of $U \oplus V$. Conversely, suppose that M is a nonzero irreducible submodule of $U \oplus V$. Then we obtain that

$$M \subseteq M_1 \oplus M_2$$

where $M_i = p_i(M)$ and $p_1 : U \oplus V \rightarrow U$ and $p_2 : U \oplus V \rightarrow V$ are canonical projections. I claim that M_i is also irreducible since by isomorphism theorem $M_i \cong M / \ker(p_i|_M)$ and $\ker(p_i|_M)$ as submodule of M must equal to zero or M , then M_i is isomorphic to zero or M meaning M_i is either the zero module or irreducible. Then we obtain that

$$\text{Soc}(U \oplus V) \subseteq \text{Soc}(U) \oplus \text{Soc}(V).$$

Therefore, the statement is true. ■

Problem 4

Let R be an arbitrary ring. Then every 2-sided ideal of the matrix ring $M_n(R)$ can be expressed in the form $M_n(I)$, where I is a 2-sided ideal of R . In particular, R is simple if and only if $M_n(R)$ is simple. [Hint: For an ideal L of $M_n(R)$, let I be the set of $(1,1)$ -entries of L . Show that I is an ideal of R and $L = M_n(I)$.]

Solution: Use notations above. First we need to show I is an ideal. For an arbitrary element $r \in R$, let $\text{diag}(r)$ denote diagonal matrix setting diagonal elements to be r . By our definition for any $s \in I$, there exists $A_s \in L$ such that the $(1,1)$ -entry of A_s is s . Since L is an ideal of $M_n(R)$, we obtain that

$$\text{diag}(r) \cdot A_s \in L \text{ and } A_s \cdot \text{diag}(r) \in L,$$

which implies that $r \cdot s \in I$ and $s \cdot r \in I$. Furthermore, for any $s_1, s_2 \in I$, there exist $A_{s_1}, A_{s_2} \in L$ whose $(1,1)$ -entries are s_1 and s_2 , respectively. Since L is an additive subgroup, $A_{s_1} - A_{s_2} \in L$, and its $(1,1)$ -entry is $s_1 - s_2$, meaning $s_1 - s_2 \in I$. Thus, I is an ideal. Next, we need to prove that $L = M_n(I)$. Let E_{ij} denote the matrix whose (i,j) -entry is 1 and other entries are 0. $\{E_{ij}\}$ form a basis of $M_n(R)$. Suppose that B is an arbitrary matrix in L . Then we can suppose that

$$B = \sum_{i,j} b_{ij} E_{ij}$$

Since L is an ideal then by

$$E_{1k} \cdot B \cdot E_{l1} = b_{kl} E_{11} \in L$$

we know that arbitrary entry b_{kl} is contained in I . Therefore, we obtain that $L \subseteq M_n(I)$. Conversely, suppose B is an arbitrary matrix in $M_n(I)$. Then for (i,j) -entry of B , denoted by b_{ij} , there exists a matrix $A_{ij} \in L$ such that the $(1,1)$ -entry of A_{ij} is b_{ij} . Then by

$$B = \sum_{i,j} E_{i,1} \cdot A_{ij} \cdot E_{1,j}$$

we obtain that $M_n(I) \subseteq L$. Therefore, $M_n(R)$ has no proper ideals if and only if R has no proper ideals. Equivalently, $M_n(R)$ is simple if and only if R is simple. ■

Problem 5

If W is a submodule of a semisimple module V , then W and V/W are also semisimple.

Solution: We first show that for an arbitrary submodule of W , it is a direct summand of W and thus W is semisimple. Let U be an arbitrary submodule of W . Since $W \subseteq V$, U is also a submodule of V . Because V is semisimple, U is a direct summand of V . Thus, there exists a submodule U' of V such that

$$V = U \oplus U'.$$

Intersecting both sides with W , we get

$$W = V \cap W = (U \oplus U') \cap W.$$

Since $U \subseteq W$, we can apply Dedekind's modular law to obtain

$$W = U \oplus (U' \cap W).$$

Note that $U' \cap W$ is a submodule of W . This shows that any submodule U of W is a direct summand of W .

Next, we show that V/W is semisimple. Since W is a submodule of the semisimple module V , it is a direct summand of V . Therefore, there exists a submodule W' of V such that

$$V = W \oplus W'.$$

By the First Isomorphism Theorem, we have

$$V/W = (W \oplus W')/W \cong W'.$$

Since W' is a submodule of the semisimple module V , by the conclusion from above, W' is also semisimple. Because V/W is isomorphic to a semisimple module W' , it follows that V/W itself is semisimple. ■

8 Week 6

Problem 1

Let D be a division ring and n a positive integer.

- (a) Show that the natural left $M_n(D)$ -module, consisting of column vectors of length n with entries in D , is a simple module.
- (b) Show that $M_n(D)$ is semisimple and has up to isomorphism only one simple module.
- (c) Show that every ring of the form

$$M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$$

is semisimple, where D_i are division rings.

- (d) Show that $M_n(D)$ is a simple ring, namely, one in which the only 2-sided ideals are the zero ideal and the whole ring.

Solution: (a) Let x be an arbitrary nonzero n dimensional column vector of D . Suppose that the i -entry of x is nonzero and denoted by x_i . Let E_{ji} denote the $n \times n$ matrix such that the (j, i) entry is the identity in D and other entries are zero. Then we obtain that

$$(x_i^{-1}E_{ji})x = \varepsilon_j$$

where ε_j denotes the column vector whose j entry is the identity in D and other entries are zero. Since $\{\varepsilon_j\}$ forms a basis, we obtain that if a submodule contains a nonzero element x , then we can use x to generate any vector. Therefore, the module of column vectors D^n is a simple $M_n(D)$ -module.

(b) Let A_j denotes the submodule of $M_n(D)$ consisting of matrices whose entries are zero except possibly in the i^{th} column. Then we obtain that

$$M_n(D) = A_1 \oplus \cdots \oplus A_n.$$

Since each $A_j \simeq D^n$ and D^n is simple by (a). Therefore, $M_n(D)$ is semisimple and has up to isomorphism only one simple module.

(c) Let R denote such a ring. I claim that each $M_{n_i}(D_i)$ is semisimple as a R -module. Without loss of generality, we prove it for $M_{n_1}(D_1)$. By definition, we know

$$\text{Ann}_R(M_{n_1}(D_1)) = M_{n_2}(D_2) \oplus \cdots \oplus M_{n_r}(D_r).$$

Let I denote this annihilator. Since $R/I \simeq M_{n_1}(D_1)$, $M_{n_1}(D_1)$ is semisimple as R/I -module by (b). According the discussion in our class, $M_{n_1}(D_1)$ is semisimple as R/I -module if and only if as R -module. Then each $M_{n_i}(D_i)$ is semisimple as R -module, hence R is semisimple.

(d) We use a conclusion from Homework 6: Every 2-sided ideal of the matrix ring $M_n(D)$ can be expressed in the form $M_n(I)$ where I is a 2-sided ideal of D . Since D is a division ring, the 2-sided ideals of D are trivial. Therefore, $M_n(D)$ is a simple ring. ■

Problem 2

Let U be an A -module for a semisimple finite-dimensional algebra A (over a field). Show that if $\text{End}_A(U)$ is a division ring then U is simple.

Solution: Since A is a semisimple finite-dimensional algebra, every A -module is semisimple. Therefore, we can suppose that

$$U \simeq S_1 \oplus \cdots \oplus S_k.$$

The endomorphism ring is then given by $\text{End}_A(U) \cong \bigoplus_{p,q} \text{Hom}_A(S_p, S_q)$. If U contained two non-isomorphic simple modules S_i and S_j , $\text{End}_A(U)$ would have a non-trivial idempotent corresponding to the projection onto one of the components, contradicting the fact that $\text{End}_A(U)$ is a division ring.

Thus, all simple components must be isomorphic to some simple module S , meaning $U \cong S^k$. According to the properties of endomorphism rings of semisimple modules:

$$\text{End}_A(U) \cong M_k(\text{End}_A(S))$$

By Schur's Lemma, $D' = \text{End}_A(S)$ is a division ring. Therefore, $\text{End}_A(U) \cong M_k(D')$. A matrix ring $M_k(D')$ over a division ring is itself a division ring if and only if $k = 1$. Since if dimension greater than one, there exists non-invertible matrices. Consequently, we have

$$U \simeq S$$

showing that U is simple. ■

Problem 3

Let A be a ring. Then A is a semisimple Artinian ring if and only if every A -module is projective.

Solution: Suppose that A is a semisimple Artinian ring. We want to show that every A -module is projective. For any arbitrary left A -module N , there exists a free A -module F and an epimorphism $f : F \rightarrow N$. Since the free module F is a direct sum of copies of semisimple module ${}_A A$, it follows that F is also a semisimple module. Therefore, the kernel of the epimorphism $\ker(f)$ is a direct summand of F . This means the short exact sequence

$$0 \rightarrow \ker(f) \rightarrow F \xrightarrow{f} N \rightarrow 0$$

splits. Consequently, $F \cong \ker(f) \oplus N$, which implies that N is isomorphic to a direct summand of the free module F meaning that N is projective.

Conversely, suppose that every left A -module is projective. We want to show that A is a semisimple Artinian ring. Let I be an arbitrary left ideal of A . The quotient A/I is naturally a left A -module. Consider the canonical short exact sequence:

$$0 \rightarrow I \rightarrow A \xrightarrow{\pi} A/I \rightarrow 0$$

where π is the natural projection. By our assumption, every A -module is projective, so A/I is a projective module and thus the short exact sequence splits. We obtain that

$$A \simeq I \oplus A/I.$$

Therefore every submodule of A splits A , we obtain that A is semisimple. Since A is semisimple ring, it is also Artinian. To see that, We may suppose that

$${}_A A = \bigoplus_{i \in I} S_i$$

where S_i is simple left ideal. Suppose that $1 = e_1 + \cdots + e_n$ with $e_i \in S_i$ and therefore

$${}_A A = S_1 \oplus S_2 \oplus \cdots \oplus S_n$$

implying its Artinian. ■

Problem 4

Let k be a field. Suppose the group algebra kG is semisimple, and the simple kG -modules are $S_1, \dots, S_r$ with degrees $d_i = \dim_k S_i$. Show that

$$\sum_{i=1}^r d_i^2 \geq |G|$$

with equality if and only if $\text{End}_{kG}(S_i) = k$ for all i .

Solution: According to Wedderburn-Artin theorem, we can suppose that

$$kG \cong M_{n_1}(D_1) \oplus M_{n_2}(D_2) \oplus \cdots \oplus M_{n_r}(D_r).$$

Hence, we can suppose that $M_{n_i}(D_i) \simeq S_i^{n_i}$. Since $M_{n_i}(D_i)$ only consist of one simple module up to isomorphism and each simple module can be interpreted as the submodule of kG . Then

for each simple module S_j , it can only be contained in one component in our decomposition meaning that our assumption is valid. Therefore, we may assume that

$$kG \cong S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}.$$

Hence we obtain that

$$|G| = \dim_k(kG) = \sum_{i=1}^r n_i d_i.$$

Since S_i as a simple submodule of $M_{n_i}(D_i)$, then we can say $S_i \simeq D_i^{n_i}$ as kG -module isomorphism. According to Schur's lemma, $\text{End}_{kG}(S_i)$ must be a division ring. It means that $\text{End}_{kG}(S_i)$ only consists of the matrices of the form dI_{n_i} where $d \in D_i$. Hence we can write

$$\text{End}_{M_{n_i}(D_i)}(S_i) \simeq D_i^{\text{op}}.$$

We can rewrite our formula of dimension by

$$n_i = \dim_k(S_i) / \dim_k(D_i).$$

We obtain that

$$|G| = \sum_{i=1}^r d_i^2 / \dim_k(D_i).$$

Therefore, the equality holds if and only if $\dim_k(\text{End}_{M_{n_i}(D_i)}(S_i)) = \dim_k(D_i) = 1$. It also equivalently say $\text{End}_{M_{n_i}(D_i)}(S_i) \simeq k$. ■

Problem 5

Suppose that we have R -module homomorphisms $U \xrightarrow{f} V \xrightarrow{g} W$. Show that: if gf is an essential epimorphism and f is an epimorphism, then both f and g are essential epimorphisms.

Solution: Since gf is an epimorphism, g must be an epimorphism. Suppose that f is not essential. Equivalently, there exists a proper submodule N of U such that $f|_N$ is an epimorphism. Since g is an epimorphism, we now obtain that

$$g \circ f|_N : N \rightarrow W$$

is also an epimorphism. However, since gf is essential, we obtain that $N = U$, which is a contradiction. Therefore, f is essential.

Suppose that g is not essential. Then we can find a proper submodule M of V such that $g(M) = W$. Then we consider the submodule $Q = f^{-1}(M)$ of U . It is proper, since $f(Q) = M \subsetneq V$. Then we obtain that $gf(Q) = W$. Because gf is essential, we find that $Q = U$, which is a contradiction. Therefore g must be essential. Our claim holds. ■

9 Week 7

Problem 1

Let V be an A -module.

- (a) Show that an endomorphism $e : V \rightarrow V$ is a projection onto a A -submodule W if and only if e is idempotent as an element of $\text{End}_A(V)$.
- (b) Show that direct sum decompositions $V = W_1 \oplus W_2$ as A -modules are in bijection with idempotent decompositions $1 = e + f$ in $\text{End}_A(V)$.
- (c) Show that $e \in \text{End}_A(V)$ is primitive if and only if $e(V)$ is indecomposable.
- (d) Suppose that V is semisimple with finitely many simple summands and let $e_1, e_2 \in \text{End}_A(V)$ be idempotents. Show that $e_1(V) \cong e_2(V)$ as A -modules if and only if e_1 and e_2 are conjugate by an invertible element of $\text{End}_A(V)$.
- (e) Let k be a field. Show that all primitive idempotent elements in $M_n(k)$ are conjugate under the action of the unit group $GL_n(k)$, which consists of all the invertible matrices in $M_n(k)$.

Solution: (a) If e is a projection, by definition, we can suppose that $V = W \oplus W'$ such that $e(W') = 0$ and for an arbitrary element $w \in W$ we have $e(w) = w$. Therefore, let v be an arbitrary element in V with unique decomposition $v = w + w'$ where $w \in W$ and $w' \in W'$, then we obtain that $e^2(v) = e^2(w + w') = e(w) = w$, which implies $e^2(v) = e(v)$ and thus e is idempotent. Conversely, let $f = 1 - e$, I claim that $V = \text{im}(e) \oplus \text{im}(f)$. Then for an arbitrary element v we obtain that $v = e(v) + f(v)$ and thus $V = \text{im}(e) + \text{im}(f)$. Let $v \in \text{im}(e) \cap \text{im}(f)$, it means that there exists an element v' such that $v = f(v')$. Since e is idempotent, we then have $v = e(v) = ef(v') = e(1 - e)(v') = 0$. Therefore we obtain that $V = \text{im}(e) \oplus \text{im}(f)$ and e act as an identity on $\text{im}(e)$ and as null morphism on $\text{im}(f)$, which implies that e is a projection.

(b) Let Dirsum and Idmdec denotes the set of direct sum decompositions and the set of idempotent decompositions respectively. The process of (a) have shown that $\text{Idmdec} \rightarrow \text{Dirsum}$. Conversely, suppose that we have a direct sum decomposition $V = W_1 \oplus W_2$. Then we let e and f be the canonical projection to W_1 and W_2 respectively. Clearly, $e + f = 1$ satisfies our requirement. To show bijectivity, given that an idempotent morphism e . Let $f = 1 - e$, we obtain that $V = \text{im}(e) \oplus \text{im}(f)$ according to (a). Then e and f are identified as the canonical projection to $\text{im}(e)$ and $\text{im}(f)$ respectively. It means that $\text{Idmdec} \rightarrow \text{Dirsum} \rightarrow \text{Idmdec}$ is an identity. Hence, given that direct sum decomposition $V = W_1 \oplus W_2$, let e and f denotes the canonical projection to W_1 and W_2 respectively. According to the definition of canonical projection, we know that e is idempotent and $e + f = 1$. Hence we have $W_1 = \text{im}(e)$ and $W_2 = \text{im}(f)$. It means that $\text{Dirsum} \rightarrow \text{Idmdec} \rightarrow \text{Dirsum}$ is an identity. Therefore $\text{Dirsum} \rightarrow \text{Idmdec}$ is a bijection.

(c) Let W denote $e(V)$. If W is decomposable, according to (b) then there exists an idempotent decomposition $e|_W = \text{id}|_W = \varphi + \psi$. Let $i : W \rightarrow V$ be the canonical inclusion by considering the direct sum decomposition induced by idempotent decomposition $\text{id}|_V = e + (1 - e)$. Let $\tilde{\varphi} = i \circ \varphi \circ e$ and $\tilde{\psi} = i \circ \psi \circ e$. Both $\tilde{\varphi}$ and $\tilde{\psi}$ are an idempotent morphism on V . Clearly both of them act as null morphism on $\text{im}(1 - e) = \ker(e)$ and for an arbitrary element $v \in \text{im}(e)$, we have $\tilde{\varphi}^2(v) = i \circ \varphi \circ e \circ i \circ \varphi \circ e(v) = i \circ \varphi^2 \circ e(v) = \tilde{\varphi}(v)$. Simlilar for $\tilde{\psi}$. Therefore, we obtain idempotent decomposition $e = \tilde{\varphi} + \tilde{\psi}$. Conversely, suppose that e is not primitive, then there exists an idempotent decomposition $e = \varphi + \psi$. Both φ and ψ act idempotently on W . Therefore we obtain an idempotent decomposition on W which is $e|_W = \text{id}|_W = \varphi|_W + \psi|_W$. According to (b), it corresponds to a direct sum decomposition of W . Therefore, our claim holds.

(d) Suppose that e_1 and e_2 are conjugate by a . Namely, we assume that $e_1 = ae_2a^{-1}$ it gives an isomorphism $e_2(V) \rightarrow e_1(V)$ by $v \mapsto a(v)$. We just need to verify its well-definedness. Let v be an arbitrary element of $e_2(V)$, it means that there exists an element $v' \in V$ such that $v = e_2(v')$. Since a is invertible, then we obtain that $v = e_2 \circ a^{-1} \circ a(v')$. By our assumption,

then we have $a(v) = e_1 \circ a(v') \in e_1(V)$. Since a is invertible on V , the morphism $e_2(V) \rightarrow e_1(V)$ is an isomorphism. Conversely, suppose there exists an isomorphism $a : e_2(V) \rightarrow e_1(V)$. I claim that there exists a morphism $\tilde{a}$ such that $\tilde{a}|_{e_2(V)} = a$ and $e_1 = \tilde{a}e_2\tilde{a}^{-1}$. Since V is semisimple, then we can assume that

$$V \simeq e_1(V) \oplus W_1 \simeq e_2(V) \oplus W_2.$$

where W_1 and W_2 are the complement of V_1 and V_2 induced by idempotent morphisms e_1 and e_2 respectively. Because $e_1(V) \simeq e_2(V)$, we have $W_1 \simeq W_2$ by considering the simple module decomposition $V \cong \bigoplus_{i=1}^k S_i^{n_i}$. Therefore, we let $b : W_2 \rightarrow W_1$ be an arbitrary isomorphism and let $\tilde{a} = a \oplus b$, we find that $\tilde{a}|_{e_2(V)} = a$. To see the equation $e_1 = \tilde{a}e_2\tilde{a}^{-1}$ holds, by given an arbitrary element $v \in V$, there exists decomposition $v = v_2 + w_2$ and $\tilde{a}(v) = a(v_2) + b(w_2)$. Then we obtain that

$$(e_1 \circ \tilde{a} - \tilde{a} \circ e_2)(v) = e_1(a(v_2) + b(w_2)) - \tilde{a}(v_2) = a(v_2) - a(v_2) = 0.$$

Our claim holds.

(e) Let $V = k^n$ be a k -module. According to Artin–Wedderburn theorem, we obtain that $\text{End}_k(V) = M_n(D)$ where $D = \text{End}_k(k) = k$. Given that two arbitrary primitive idempotent elements e_1 and e_2 in $M_n(k)$, we must have $e_1(V) \simeq e_2(V) \simeq k$ according to (c). Hence, by (d), we obtain that e_1 and e_2 are conjugate by an invertible element of $M_n(k)$. Therefore, all primitive idempotent elements are conjugate under the action of unit group $\text{GL}_n(k)$. ■

Problem 2

Prove: No non-zero idempotent e of A is contained in $J(A)$. [Hint: $e(1 - e) = 0$.]

Solution: Suppose that there exists a non-zero idempotent element e in $J(A)$. Then we know that $1 - e$ is invertible. Therefore, we can reduce $1 - e$ on each sides of equation $e(1 - e) = 0$, we know obtain that $e = 0$ is trivial contradicting our assumption. ■

Problem 3

Prove: Let A be an Artinian ring with Artinian center $Z(A)$. Then $J(Z(A)) = J(A) \cap Z(A)$.

Solution: Suppose that $x \in J(A) \cap Z(A)$. Then for arbitrary element $y \in Z(A)$, we obtain that $1 - yx$ is invertible in A . Since $1 - yx$ is commutative and therefore its inverse is included in $Z(A)$. It means that $x \in J(Z(A))$. Conversely, since A and $Z(A)$ are Artinian, their Jacobson radicals are nilpotent ideals. Therefore, let x be a nilpotent element in $Z(A)$, it is also nilpotent in A . Of course, it is commutative. Therefore we obtain $J(Z(A)) \subseteq J(A) \cap Z(A)$. ■

Problem 4

Prove: Let $A = A_1 \oplus \cdots \oplus A_n$ be a direct sum of 2-sided ideals. Then

$$(i) \quad Z(A) = Z(A_1) \oplus Z(A_2) \oplus \cdots \oplus Z(A_n).$$

$$(ii) \quad J(A) = J(A_1) \oplus J(A_2) \oplus \cdots \oplus J(A_n).$$

Solution: (i) Let $x \in Z(A)$. It admits a decomposition $x = \sum_i x_i$ where $x_i \in A_i$. Since A_i is double sided ideals, then we obtain that $A_i A_j \subseteq A_i \cap A_j = 0$ and $A_j A_i = 0$. Therefore, for arbitrary $y_i \in A_i$, the equation $xy_i = y_i x$ implies that $x_i y_i = y_i x_i$. It means that $x_i \in Z(A_i)$ and thus $Z(A) \subseteq \bigoplus_i Z(A_i)$. Conversely, suppose that $x_i \in Z(A_i)$ and $x = \sum_i x_i$ is clearly commutative. Then $Z(A) \supseteq \bigoplus_i Z(A_i)$ and thus the claim holds.

(ii) Given that $x \in J(A)$ with decomposition $x = \sum_i x_i$ and $y_i \in A_i$. We find that $1 - y_i x = 1 - y_i x_i$ is invertible in A . Suppose that $u(1 - y_i x) = 1$ and u admits a decomposition $u = \sum_i u_i$. Suppose 1 admits a decomposition $1 = \sum_i e_i$, then we obtain that $u_i(e_i - y_i x_i) = e_i$. Therefore, $e_i - y_i x_i$ is invertible in A_i and thus $J(A) \subseteq \bigoplus_i J(A_i)$. Conversely, given that $x_i \in J(A_i)$ and an arbitrary element $y = \sum_i y_i \in A$. We consider that $1 - yx = \sum_i (e_i - y_i x_i)$. Since $x_i \in J(A_i)$, we can suppose there exists $u_i \in A_i$ such that $u_i(e_i - y_i x_i) = e_i$. Therefore, we obtain that $u(1 - yx) = \sum_i u_i(1 - y_i x_i) = \sum_i e_i = 1$ and thus $J(A) \supseteq \bigoplus_i J(A_i)$. ■

Problem 5

Let $\phi : U \rightarrow V$ be a homomorphism of A -modules. Show that $\phi(\text{Soc } U) \subseteq \text{Soc } V$, and that if ϕ is an isomorphism then ϕ restricts to an isomorphism $\text{Soc } U \rightarrow \text{Soc } V$.

Solution: I claim that $\phi(\text{Soc}(U))$ is semisimple. Since $\text{Soc}(U)$ is semisimple, we suppose that $\text{Soc}(U) = \bigoplus_{i \in I} U_i$. Since U_i is simple, then $\phi(U_i)$ must be zero module or isomorphic to U_i . In any case, $\phi(U_i)$ must be either zero or simple. $\phi(\text{Soc}(U)) \subseteq \sum_i \phi(U_i)$ implies that $\phi(\text{Soc}(U))$ as a submodule of semisimple module $\sum_i \phi(U_i)$ is also semisimple. It means that $\phi(\text{Soc}(U)) \subseteq \text{Soc}(V)$. Since ϕ is an isomorphism, for any non zero simple submodule U_i , then $\phi(U_i)$ is not zero submodule. For arbitrary two simple module U_i and U_j , we then have $\phi(U_i) \cap \phi(U_j) = \{0\}$. Therefore we can suppose that

$$\text{Soc}(U) = \bigoplus_{i \in I} U_i \text{ and } \phi(\text{Soc}(U)) = \bigoplus_{i \in I} \phi(U_i).$$

Let $\text{Soc}(V) = \phi(\text{Soc}(U)) \oplus \bigoplus_{j \in J} V_j$. By (i), we obtain that

$$\phi^{-1}(\text{Soc}(V)) = \text{Soc}(U) \oplus \bigoplus_{j \in J} \phi^{-1}(V_j) \subseteq \text{Soc}(U).$$

It means $V_j \simeq \phi^{-1}(V_j) \simeq (0)$. Equivalently, we obtain that $\text{Soc } U \rightarrow \text{Soc } V$ is an isomorphism. ■

Problem 6

If $\varphi : A \rightarrow A'$ is an epimorphism of rings, then $\varphi(J(A)) \subseteq J(A')$.

Solution: By definition, we obtain that

$$\varphi(J(A)) = \varphi \left(\bigcap_{\mathfrak{m} \in \text{Spm}(A)} \mathfrak{m} \right) \subseteq \bigcap_{\mathfrak{m} \in \text{Spm}(A)} \varphi(\mathfrak{m}).$$

where Spm denotes the spectrum of maximal ideals. Since φ is surjective, it follows that every maximal ideal $\mathfrak{m}'$ in A' corresponds to a maximal ideal $\mathfrak{m}$ in A such that $\varphi(\mathfrak{m}) = \mathfrak{m}'$. Therefore,

we obtain that

$$\varphi(J(A)) \subseteq \bigcap_{\mathfrak{m}' \in \text{Spm}(A')} \mathfrak{m}' = J(A').$$

■

10 Week 8

Problem 1

Let A be a finite-dimensional algebra over a field. Suppose that V is a f.g. A -module, and that a certain simple A -module S occurs as a composition factor of V with multiplicity 1. Suppose that there exist nonzero homomorphisms $S \rightarrow V$ and $V \rightarrow S$. Prove that S is a direct summand of V .

Solution: Let φ and ψ denote nonzero homomorphism $S \rightarrow V$ and $V \rightarrow S$ respectively. Since S is simple, we obtain that φ must be injective. Otherwise, it should be a zero homomorphism which contradicts our conditions. For the same reason, we obtain that ψ must be surjective. Hence, I claim that $\psi \circ \varphi \in \text{End}_A(S)$ is an isomorphism or a zero homomorphism according to Schur's Lemma. Suppose that $\psi \circ \varphi = 0$. We obtain that a descending chain of submodules of V

$$V \supseteq \ker \psi \supseteq \text{im } \varphi \supseteq (0).$$

Since φ is injective, we obtain that $S \simeq \text{im } \varphi$. By the First Isomorphism Theorem for modules, we also have $S \simeq V/\ker \psi$. It implies that S occurs twice as a composition factor which is a contradiction. Therefore, $\psi \circ \varphi$ must be an isomorphism. Let $f = (\psi \circ \varphi)^{-1}$ then we obtain that $\psi \circ (\varphi \circ f) = \text{id}_S$. It follows that the short exact sequence

$$0 \rightarrow \ker \psi \rightarrow V \xrightarrow{\psi} S \rightarrow 0$$

is split. Equivalently, we say $V \simeq S \oplus \ker \psi$.

■

Problem 2

Suppose that U , V and W are modules for a finite-dimensional algebra over a field and P_W is the projective cover of W . Assume that these modules are finite-dimensional.

- (a) Show that $U \rightarrow W$ is an essential epimorphism if and only if there is a surjective homomorphism $P_W \rightarrow U$ so that the composite $P_W \rightarrow U \rightarrow W$ is a projective cover of W . In this situation, show that $P_W \rightarrow U$ must be a projective cover of U .
- (b) Prove the following “extension and converse” to Nakayama's Lemma: let V be any submodule of U . Then $U \rightarrow U/V$ is an essential epimorphism if and only if $V \subseteq \text{Rad } U$.

Solution: (a) Let f denote the essential epimorphism $U \rightarrow W$. Let g denote $P_W \rightarrow W$. According to the universal property of projective module, there exists a morphism $h : P_W \rightarrow U$ such that $g = f \circ h$. Then we can see h is the homomorphism that we want. According to discussion in our class, both f and g are essential epimorphisms implies that h is also an essential epimorphism. It follows that $h : P_W \rightarrow U$ is a projective cover. Conversely, since g

is an essential epimorphism and h is a surjection, according to our homework, we obtain that $f : U \rightarrow W$ and $h : P_W \rightarrow U$ are also essentially epimorphic. It follows that $P_W \rightarrow U$ is a projective cover and $U \rightarrow W$ is an essential epimorphism.

(b) Let f denote morphism $U \rightarrow U/V$. Suppose that $V \not\subseteq \text{Rad } U$, then there exists a maximal submodule M such that $M + V = U$. By isomorphism Theorem of module, we obtain that $U/V \simeq M/(M \cap V)$. It implies that

$$f|_M : M \rightarrow U/V \simeq M/(M \cap V)$$

is an epimorphism. Equivalently, f is not an essential epimorphism.

Suppose that $V \subseteq \text{Rad } U$. Suppose that f is not essentially epimorphic, equivalently, there exists a proper submodule M of U such that $f|_M$ is epimorphic. For an arbitrary element $x \in U$, $f|_M$ is epimorphic implies that there exists $m \in M$ and $v \in V$ such that $x = m + v$. It means that $M + V = U$. However, by Nakayama's Lemma, since $V \subseteq \text{Rad } U$, it implies that $M = U$. It contradicts our assumption. Therefore, f must be essentially epimorphic. ■

Remark 10.1. There are some facts:

- Let $\text{Idem}(R) = \{e \in R \mid e^2 = e\}$ be the set of idempotents of a ring R , and let $\text{Summ}(M)$ be the set of all direct summands of M . There is a bijection:

$$\Phi : \text{Idem}(\text{End}_A(M)) \xrightarrow{\sim} \text{Summ}(M)$$

$$e \mapsto eM$$

- Let $f : M \rightarrow N$ be an essential epimorphism ($\text{Ker}(f) \subseteq \text{rad}(M)$). We define

$$\mathcal{L}_N = \{\bar{e} \in \text{Idem}(\text{End}_A(N)) \mid \exists e \in \text{Idem}(\text{End}_A(M)) \text{ s.t. } f \circ e = \bar{e} \circ f\}.$$

then there exists bijective correspondence Ψ maps these idempotents to the set of essential covers of summands:

$$\Psi : \mathcal{L}_N \xrightarrow{\sim} \{f|_{eM} : eM \rightarrow \bar{e}N \mid e \in \text{Idem}(\text{End}_A(M)) \text{ is a lift of } \bar{e}\}$$

$$\bar{e} \mapsto (f|_{eM} : eM \rightarrow \bar{e}N)$$

- A ring R is semiperfect if every finitely generated (left) R -module admits a projective cover. It is equivalent to the following conditions:
 - a The quotient $R/J(R)$ is a semisimple Artinian ring;
 - b Idempotents lift modulo $J(R)$.
- Let $\text{Top}(M) = M/\text{rad}(M)$ be its radical quotient. A module M admits a projective cover if and only if $\text{Top}(M)$ admits a projective cover.

Problem 3

The following holds for a valuation v of K :

- (i) $v(\pm 1) = 0$
- (ii) $v(a) = v(-a)$
- (iii) $v(a^{-1}) = -v(a)$
- (iv) $v(b) < v(a) \Rightarrow v(a + b) = v(b)$.

Solution: (i) Since $v(1) = v(1^n) = nv(1)$ for any integer n . For equation holds, $v(1) = 0$. Similarly, $v(-1) = v((-1)^{2n+1}) = (2n+1)v(-1)$ for any integer n , then we obtain that $v(-1) = 0$. (ii) Since (i), we obtain that $v(-a) = v(-1) + v(a) = v(a)$. (iii) Consider that $0 = v(1) = v(a \cdot a^{-1}) = v(a) + v(a^{-1})$, then $v(a^{-1}) = -v(a)$. (iv) We note that by the strongly triangular inequality, we obtain that $v(a+b) \geq \min\{v(a), v(b)\} = v(b)$. By the same reason, we consider that

$$v(b) = v(a+b-a) \geq \min\{v(a+b), v(a)\}.$$

Since $v(b) < v(a)$, then it implies that $v(b) \geq v(a+b)$ for the inequality holds. Therefore, we claim that $v(a+b) = v(b)$ if $v(b) < v(a)$. ■

11 Week 9

Problem 1

Let φ be a multiplicative valuation of a field K . Then the following holds:

- (i) (a_n) is a Cauchy sequence $\Leftrightarrow \lim_{n \rightarrow \infty} \varphi(a_{n+1} - a_n) = 0$.
- (ii) If $\lim_{n \rightarrow \infty} a_n = a \neq 0$, then $\varphi(a_n) = \varphi(a)$ for a sufficiently large n .

Solution: (i) Suppose that (a_n) is a Cauchy sequence. Then for any $\varepsilon > 0$, there exists $N \in \mathbb{Z}_{>0}$ such that

$$\varphi(a_m - a_n) < \varepsilon \text{ for any } m, n > N.$$

We take $m = n + 1$, then we obtain that $\varphi(a_{n+1} - a_n) < \varepsilon$ for any $n > N$. Equivalently, we can say $\lim_{n \rightarrow \infty} \varphi(a_{n+1} - a_n) = 0$. Conversely, suppose that $\varphi(a_{n+1} - a_n) = 0$, it means that for any $\varepsilon > 0$ there exists a positive integer N such that for any $n > N$ we have

$$\varphi(a_{n+1} - a_n) < \varepsilon.$$

Consider the strongly triangular inequality, then we have

$$\varphi(a_{n+m} - a_n) = \varphi\left(\sum_{i=0}^{m-1} (a_{n+i+1} - a_{n+i})\right) = \max_i \{\varphi(a_{n+i+1} - a_{n+i})\} < \varepsilon.$$

Therefore, we obtain that (a_n) is a Cauchy sequence.

- (ii) Since $a_n - a$ converge to zero, then there exists $N > 0$ such that for any $n > N$ we have

$$\varphi(a_n - a) < \varphi(a)/2.$$

Similarly, since a_n converge to a , then there exists $M > 0$ such that for any $n > M$ we have

$$\varphi(a_n) > \varphi(a)/2.$$

We can choose $N \geq M$, then we have $\varphi(a_n) > \varphi(a_n - a)$. It implies that

$$\varphi(a) = \varphi(a - a_n + a_n) = \max\{\varphi(a - a_n), \varphi(a_n)\} = \varphi(a_n).$$

Our claim holds. ■

Problem 2

Let φ be a multiplicative valuation of a field K . Let V be a finite dimensional K -space with basis $u_1, \dots, u_n$. For $v = a_1u_1 + \dots + a_nu_n \in V$, define $\|v\|_0$ by

$$\|v\|_0 = \max\{\varphi(a_i); 1 \leq i \leq n\}.$$

Then this is a norm on V . Moreover, the following holds concerning the metric induced by this norm.

- (i) Let $v_r = a_{r1}u_1 + \dots + a_{rn}u_n \in V$ be given for $r = 1, 2, \dots$. Then the sequence (v_r) is a Cauchy sequence if and only if the sequence $(a_{ri})_r$ in K is so for all i . Also $\lim_{r \rightarrow \infty} v_r = \sum_{i=1}^n a_i u_i$ if and only if $\lim_{r \rightarrow \infty} a_{ri} = a_i$ for all i .
- (ii) If K is complete, then so is V .

Solution: (i) Suppose that $(v_r)_r$ is a Cauchy sequence. It means that for any $\varepsilon > 0$, there exists $N > 0$ such that for any $r > N$ and index i we have

$$0 \leq \varphi(a_{r+1,i} - a_{ri}) \leq \|v_{r+1} - v_r\|_0 < \varepsilon.$$

It implies that $\lim_{r \rightarrow \infty} \varphi(a_{r+1,i} - a_{ri}) = 0$. According to Problem 1.(i), we then have $(a_{ri})_r$ is a Cauchy sequence for each i .

Conversely, suppose that $(a_{ri})_r$ is a Cauchy sequence. Then for every $\varepsilon > 0$, there exists an integer N such that for any $r > N$, an integer m and index i satisfy

$$\varphi(a_{r+m,i} - a_{r,i}) < \varepsilon.$$

Hence, we have

$$\|v_{r+m} - v_r\|_0 = \max_i \varphi(a_{r+m,i} - a_{r,i}) < \varepsilon.$$

It shows $(v_r)_r$ is a Cauchy sequence.

For next statement, it is suffice to show that $v_r \rightarrow 0$ iff. $a_{ri} \rightarrow 0$. Because we just let $w_r = v_r - \sum a_i u_i$, and the statement $w_r \rightarrow 0$ iff. $(a_{ri} - a_i) \rightarrow 0$ is equivalent to the statement we need to prove. Suppose that v_r converge to zero. It means that for any $\varepsilon > 0$, we have $N > 0$ and for any $r > N$ and index i the inequality

$$0 \leq \varphi(a_{ri}) \leq \|v_r\|_0 < \varepsilon$$

Which implies that $\varphi(a_{ri}) \rightarrow 0$ and $a_{ri} \rightarrow 0$ follows. Conversely, $a_{ri} \rightarrow 0$ implies $\varphi(a_{ri}) \rightarrow 0$ and thus we have $\|v_r\|_0 \rightarrow 0$. And $v_r \rightarrow 0$ follows. Therefore, our claim holds.

(ii) Let $(v_r)_r$ be an arbitrary Cauchy sequence. According to (i), it is equivalent to $(a_{ri})_r$ is a Cauchy sequence for each i . Since K is complete, it means that there exists $a_i \in K$ such that $(a_{ri} - a_i)_r$ converge to zero. According to our discussion above, it is equivalent to $(v_r - \sum a_i u_i)_r$ converge to zero. Therefore, V is complete as well. ■

12 Week 10

Problem 1

Let φ be a multiplicative valuation of a field K . Then the following holds:

- (i) (a_n) is a Cauchy sequence $\Leftrightarrow \lim_{n \rightarrow \infty} \varphi(a_{n+1} - a_n) = 0$.
- (ii) If $\lim_{n \rightarrow \infty} a_n = a \neq 0$, then $\varphi(a_n) = \varphi(a)$ for a sufficiently large n .

Solution: (i) Suppose that (a_n) is a Cauchy sequence. Then for any $\varepsilon > 0$, there exists $N \in \mathbb{Z}_{>0}$ such that

$$\varphi(a_m - a_n) < \varepsilon \text{ for any } m, n > N.$$

We take $m = n + 1$, then we obtain that $\varphi(a_{n+1} - a_n) < \varepsilon$ for any $n > N$. Equivalently, we can say $\lim_{n \rightarrow \infty} \varphi(a_{n+1} - a_n) = 0$. Conversely, suppose that $\varphi(a_{n+1} - a_n) = 0$, it means that for any $\varepsilon > 0$ there exists a positive integer N such that for any $n > N$ we have

$$\varphi(a_{n+1} - a_n) < \varepsilon.$$

Consider the strongly triangular inequality, then we have

$$\varphi(a_{n+m} - a_n) = \varphi\left(\sum_{i=0}^{m-1} (a_{n+i+1} - a_{n+i})\right) = \max_i \{\varphi(a_{n+i+1} - a_{n+i})\} < \varepsilon.$$

Therefore, we obtain that (a_n) is a Cauchy sequence.

(ii) Since $a_n - a$ converge to zero, then there exists $N > 0$ such that for any $n > N$ we have

$$\varphi(a_n - a) < \varphi(a)/2.$$

Similarly, since a_n converge to a , then there exists $M > 0$ such that for any $n > M$ we have

$$\varphi(a_n) > \varphi(a)/2.$$

We can choose $N \geq M$, then we have $\varphi(a_n) > \varphi(a_n - a)$. It implies that

$$\varphi(a) = \varphi(a - a_n + a_n) = \max\{\varphi(a - a_n), \varphi(a_n)\} = \varphi(a_n).$$

Our claim holds. ■

Problem 2

Let φ be a multiplicative valuation of a field K . Let V be a finite dimensional K -space with basis $u_1, \dots, u_n$. For $v = a_1u_1 + \dots + a_nu_n \in V$, define $\|v\|_0$ by

$$\|v\|_0 = \max\{\varphi(a_i); 1 \leq i \leq n\}.$$

Then this is a norm on V . Moreover, the following holds concerning the metric induced by this norm.

- (i) Let $v_r = a_{r1}u_1 + \dots + a_{rn}u_n \in V$ be given for $r = 1, 2, \dots$. Then the sequence (v_r) is a Cauchy sequence if and only if the sequence $(a_{ri})_r$ in K is so for all i . Also $\lim_{r \rightarrow \infty} v_r = \sum_{i=1}^n a_i u_i$ if and only if $\lim_{r \rightarrow \infty} a_{ri} = a_i$ for all i .
- (ii) If K is complete, then so is V .

Solution: (i) Suppose that $(v_r)_r$ is a Cauchy sequence. It means that for any $\varepsilon > 0$, there exists $N > 0$ such that for any $r > N$ and index i we have

$$0 \leq \varphi(a_{r+1,i} - a_{ri}) \leq \|v_{r+1} - v_r\|_0 < \varepsilon.$$

It implies that $\lim_{r \rightarrow \infty} \varphi(a_{r+1,i} - a_{ri}) = 0$. According to Problem 1.(i), we then have $(a_{ri})_r$ is a Cauchy sequence for each i .

Conversely, suppose that $(a_{ri})_r$ is a Cauchy sequence. Then for every $\varepsilon > 0$, there exists an integer N such that for any $r > N$, an integer m and index i satisfy

$$\varphi(a_{r+m,i} - a_{r,i}) < \varepsilon.$$

Hence, we have

$$\|v_{r+m} - v_r\|_0 = \max_i \varphi(a_{r+m,i} - a_{r,i}) < \varepsilon.$$

It shows $(v_r)_r$ is a Cauchy sequence.

For next statement, it is suffice to show that $v_r \rightarrow 0$ iff. $a_{ri} \rightarrow 0$. Because we just let $w_r = v_r - \sum a_i u_i$, and the statement $w_r \rightarrow 0$ iff. $(a_{ri} - a_i) \rightarrow 0$ is equivalent to the statement we need to prove. Suppose that v_r converge to zero. It means that for any $\varepsilon > 0$, we have $N > 0$ and for any $r > N$ and index i the inequality

$$0 \leq \varphi(a_{ri}) \leq \|v_r\|_0 < \varepsilon$$

Which implies that $\varphi(a_{ri}) \rightarrow 0$ and $a_{ri} \rightarrow 0$ follows. Conversely, $a_{ri} \rightarrow 0$ implies $\varphi(a_{ri}) \rightarrow 0$ and thus we have $\|v_r\|_0 \rightarrow 0$. And $v_r \rightarrow 0$ follows. Therefore, our claim holds.

(ii) Let $(v_r)_r$ be an arbitrary Cauchy sequence. According to (i), it is equivalent to $(a_{ri})_r$ is a Cauchy sequence for each i . Since K is complete, it means that there exists $a_i \in K$ such that $(a_{ri} - a_i)_r$ converge to zero. According to our discussion above, it is equivalent to $(v_r - \sum a_i u_i)_r$ converge to zero. Therefore, V is complete as well. ■

13 Midterm

Lemmas for Problem

Lemma 13.1. *A $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$ -module.*

Proof. Given an arbitrary $\mathbb{Z}$ -module G with its submodule H and a $\mathbb{Z}$ -module homomorphism $f : H \rightarrow \mathbb{Q}/\mathbb{Z}$, let

$$\Omega = \{J \supseteq H \mid \text{there exists } f_J : J \rightarrow \mathbb{Q}/\mathbb{Z} \text{ such that } f_J|_H = f\}.$$

According to Zorn's lemma, the partially ordered set Ω has a maximal element K . Suppose that K is a proper submodule of G . Choose an arbitrary element $a \in G \setminus K$, we let $\tilde{K}$ denote the submodule generated by K and a .

If the intersection $\langle a \rangle \cap K$ only contains zero, equivalently, $\tilde{K} = K \oplus \mathbb{Z} \cdot a$, then we construct a map $\tilde{f}_K : \tilde{K} \rightarrow \mathbb{Q}/\mathbb{Z}$ by sending a to zero and identified with f_K on K . Clearly, it is a $\mathbb{Z}$ -module homomorphism since $\tilde{f}_K$ can be identified with the composite of $\tilde{K} \rightarrow K \rightarrow \mathbb{Q}/\mathbb{Z}$ with the first homomorphism is the canonical projection. It is impossible because we already set K is the maximal element but we find a larger element $\tilde{K}$ with homomorphism $\tilde{f}_K$.

Next, We let the integer n to be the minimal positive integer that makes $na \in K$ holds. We construct a map $\tilde{f}_K : \tilde{K} \rightarrow \mathbb{Q}/\mathbb{Z}$ by sending a to $f_K(na)/n$ and $\tilde{f}_K|_K = f_K$. It is a well-defined module homomorphism since if we have $k - ma = 0$ in $\tilde{K}$ where $k \in K$ then it implies that $n \mid m$ otherwise there must exists a positive integer $n' < n$ such that $n \equiv n' \pmod{m}$ and then $a^{n'} \in K$ which is a contradiction. Therefore, we have $\tilde{f}_K(k) - \tilde{f}_K(ma) = f_K(k - ma) = 0$.

The process that we extend f_K to $\tilde{f}_K$ implies that K must not be proper, equivalently, $K = G$. Then we obtain that f can be extended to $G \rightarrow \mathbb{Q}/\mathbb{Z}$, it means that $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$ -module. □

Lemma 13.2. *Let A , B , and C be R -modules such that $A \subseteq B \subseteq C$. If A is an essential submodule of B and B is an essential submodule of C , then A is an essential submodule of C .*

Proof. We let $\subseteq_e$ denote essential extension. Let W be an arbitrary non-zero submodule of C . To prove that $A \subseteq_e C$, we need to show that $W \cap A \neq 0$. Since $B \subseteq_e C$ and W is a non-zero submodule of C , the intersection of W and B must be non-trivial, i.e., $W \cap B \neq 0$.

Since $A \subseteq_e B$ and $W \cap B$ is a submodule of B , we obtain that

$$(W \cap B) \cap A \neq 0$$

which implies that $W \cap A \neq 0$. Therefore, we obtain that $A \subseteq_e C$. $\square$

Lemma 13.3. *Let A and B be R -modules such that $A \subseteq_e B$. Let $f : B \rightarrow C$ be a R -module monomorphism, then $f(A) \subseteq_e f(B)$.*

Proof. Given that an arbitrary submodule W of $f(B)$. Since $A \subseteq_e B$, we have

$$f^{-1}(W) \cap A \neq 0.$$

Since f is monomorphic, we then have

$$W \cap f(A) \neq 0$$

which implies that $A \subseteq_e B$. $\square$

Problem 1

Let R be a ring with 1.

- (i) Let $h : U \rightarrow V$ be a monomorphism of R -modules. Then the following two conditions are equivalent.
 - (a) A homomorphism $k : V \rightarrow Y$ must be a monomorphism whenever the composite map $U \xrightarrow{h} V \xrightarrow{k} Y$ is a monomorphism.
 - (b) If $0 \neq W \subset V$, then $h(U) \cap W \neq 0$.

The monomorphism h that satisfy the above conditions is said to be *essential*. If there is an injective R -module Q with an essential monomorphism $h : V \rightarrow Q$, then the diagram $V \xrightarrow{h} Q$ is called an *injective hull* of V .

- (ii) Prove that: any R -module V has an injective hull.

Solution: (i) We first show that (a) implies (b). Suppose that $W \neq 0$ and $h(U) \cap W = 0$, then I claim that $U \rightarrow V/W$ is injective. Since intersection $h(U) \cap W = 0$, we obtain that

$$h(U) + W \simeq h(U) \oplus W.$$

Since h is a monomorphism, there is a canonical isomorphism $h(U) \simeq U$ and a canonical inclusion

$$U \hookrightarrow h(U) \oplus W.$$

In this sense, we find that the homomorphism $U \rightarrow V/W$ can be decomposed by

$$U \hookrightarrow h(U) \oplus W \rightarrow (h(U) \oplus W)/W.$$

The last step is exactly the canonical projection, then we obtain that $U \rightarrow V/W$ is an injection. However, the homomorphism $V \rightarrow V/W$ is a monomorphism if and only if $W = (0)$ which contradicts our assumption.

Conversely, we suppose that $k \circ h$ is a monomorphism but k is not injective. Since k is not injective, it means that $\ker k \neq 0$. Hence, the injectivity of $k \circ h$ implies that $\ker(k \circ h) \supseteq (h(U) \cap \ker k)$ and thus $h(U) \cap \ker k = 0$. It is a contradiction because we require that for an arbitrary submodule W of V we must have $h(U) \cap W \neq 0$ and $\ker k$ does not satisfy this requirement.

(ii) For a fixed R -module V , we construct an injective hull for it as follow. According to **Lemma A.**, we know that $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$ -module, it equals the functor $\text{Hom}_{\mathbb{Z}}(-, \mathbb{Q}/\mathbb{Z})$ is exact. Let $Q = \text{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})$ be the Pontryagin dual module of R . Since the tensor-hom duality, we obtain that

$$\text{Hom}_R(M, Q) \simeq \text{Hom}_{\mathbb{Z}}(M \otimes_R R, \mathbb{Q}/\mathbb{Z}) \simeq \text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$$

For arbitrary R -module M . It means that the functor $\text{Hom}_R(-, Q)$ is also exact, thus Q is an injective R -module. To construct an injective hull for R -module V , we first view V as a $\mathbb{Z}$ -module and consider the $\mathbb{Z}$ -module homomorphism

$$f : V \rightarrow \prod_{v \in V} \mathbb{Q}/\mathbb{Z}$$

determined by $\psi_v : V \rightarrow \mathbb{Q}/\mathbb{Z}$ by extending the $\mathbb{Z}$ -module homomorphism $\langle v \rangle \rightarrow \mathbb{Q}/\mathbb{Z}$ by sending v to $1/2 + \mathbb{Z}$ if $\#\langle v \rangle = \infty$ or to $1/n$ if $\#\langle v \rangle = n$. The extension exists because $\mathbb{Q}/\mathbb{Z}$ is an injective module. We also note that f is injective since for an arbitrary element $v \in \ker f$, we have $\psi_v(v) = \mathbb{Z}$. It means that the order of $\langle v \rangle$ is one, equivalently, $1 \cdot v = 0$ and thus we have $v = 0$. The injectivity of f holds.

It induces an R -module homomorphism

$$\tilde{f} : \text{Hom}_{\mathbb{Z}}(R, V) \rightarrow \text{Hom}_{\mathbb{Z}}(R, \prod_{v \in V} \mathbb{Q}/\mathbb{Z}).$$

Since we have $\text{Hom}_{\mathbb{Z}}(R, V) \simeq V$ and $\text{Hom}_R(R, \prod_{v \in V} \mathbb{Q}/\mathbb{Z}) \simeq \prod_{v \in V} Q$. We can rewrite this homomorphism

$$\tilde{f} : V \rightarrow \prod_{v \in V} Q.$$

First of all, the limit $\prod_{v \in V} Q$ is an injective R -module since given by an arbitrary R -module monomorphism $A \rightarrow B$ with homomorphism $A \rightarrow \prod_{v \in V} Q$, The second homomorphism induces homomorphisms $A \rightarrow Q$ for each index $v \in V$ by composite the canonical projection. Because Q is an injective R -module, there exists lift $B \rightarrow Q$ for each $A \rightarrow Q$. Hence, by the universal property of production, $B \rightarrow Q$ for each index induces a R -module homomorphism $B \rightarrow \prod_{v \in V} Q$ such that $A \rightarrow \prod_{v \in V} Q$ can be decomposed to $A \rightarrow B \rightarrow \prod_{v \in V} Q$. It means that $\prod_{v \in V} Q$ is an injective R -module. Hence, $\tilde{f}$ is injective since f is injective and functor $\text{Hom}_{\mathbb{Z}}(R, -)$ is left exact.

Now, we consider the partially ordered set

$$\Omega = \left\{ E \mid E \subseteq \prod_{v \in V} Q \text{ and } F \cap \tilde{f}(V) \neq 0 \text{ for any nonzero submodule } F \text{ of } E \right\}.$$

According to Zorn's lemma, Ω admits a maximal element we denote it by E . Hence, we consider the another partially ordered set

$$\Gamma = \left\{ K \mid K \subseteq \prod_{v \in V} Q \text{ and } K \cap E = 0 \right\}.$$

Similarly, we let K denote a maximal element of Γ . I claim that $\prod_{v \in V} Q \simeq K \oplus E$. To see that we consider a homomorphism $E \rightarrow (\prod_{v \in V} Q)/K$. Since $E \cap K = 0$, this homomorphism is injective. Therefore, $\prod_{v \in V} Q$ is an injective module implies that there exists a homomorphism

$$\Phi : \left(\prod_{v \in V} Q \right) / K \rightarrow \prod_{v \in V} Q$$

such that the diagram

$$\begin{array}{ccc} E & \xrightarrow{\iota} & (\prod_{v \in V} Q) / K \\ & \searrow \psi & \downarrow \Phi \\ & & \prod_{v \in V} Q \end{array}$$

commutes. We notice that Φ is injective. Because ψ is injective, then $\Phi|_{\iota(E)}$ is injective, therefore we have $\ker \Phi \cap \iota(E) = 0$. If $\ker \Phi \neq (0)$, we consider the canonical quotient

$$\pi : \prod_{v \in V} Q \rightarrow \left(\prod_{v \in V} Q \right) / K$$

and we obtain that $\pi^{-1}(\ker \Phi) \cap E \subseteq K$, since $E \cap K = 0$, then we have $\pi^{-1}(\ker \Phi) \cap E = 0$. Clearly, $\pi^{-1}(\ker \Phi) \not\subseteq K$ then we find that $(K + \pi^{-1}(\ker \Phi)) \cap E = 0$ and $K \subsetneq (K + \pi^{-1}(\ker \Phi))$. It contradicts the maximality of K , therefore Φ must be injective. The injectivity of Φ implies that $\iota(E)$ is an essential submodule of $(\prod_{v \in V} Q)/K$ since if there exists a submodule $M + K$ of $(\prod_{v \in V} Q)/K$ such that $(M + K) \cap \iota(E) = 0$, then we have $\Phi(M + K) \cap E = 0$. It implies that $\Phi(M + K) \subseteq K$ equivalently $M + K \in \ker \Phi$. Therefore, we obtain that $\iota(E)$ is an essential submodule of $(\prod_{v \in V} Q)/K$.

According to **Lemma C.**, the essential extension $\tilde{f}(V) \subseteq_e E$ implies that $\iota \circ \tilde{f}(V) \subseteq_e \iota(E)$ since ι is injective. Hence **Lemma B.** implies that

$$\iota \circ \tilde{f}(V) \subseteq_e (\prod_{v \in V} Q) / K$$

due to the fact that $\iota \circ \tilde{f}(V) \subseteq_e \iota(E)$ and $\iota(E) \subseteq_e (\prod_{v \in V} Q)/K$. The injectivity of Φ implies that

$$\tilde{f}(V) \subseteq_e \Phi \left((\prod_{v \in V} Q) / K \right).$$

Since we also have $E \subseteq \Phi((\prod_{v \in V} Q)/K)$, the maximality of E implies that

$$E = \Phi \left((\prod_{v \in V} Q) / K \right).$$

Equivalently we say $E \simeq (\prod_{v \in V} Q)/K$ and therefore

$$\prod_{v \in V} Q = K \oplus E.$$

then E as a summand of injective R -module $\prod_{v \in V} Q$ is also a injective R -module. The definition of Ω yields E satisfies condition **(i.b)**, therefore E is an injective hull of V . Our discussion implies that every R -module has an injective hull. ■

Problem 2

(i) The following holds for an idempotent e of R :

$$J(eRe) = eJ(R)e = eRe \cap J(R).$$

(ii) Prove: $J(M_n(R)) = M_n(J(R))$. Here $M_n(R)$ means the matrix ring.

Solution: (i) Suppose that $x = er_0e \in J(eRe)$, then for arbitrary $r \in R$, we consider

$$1 - rx = (1 - e) + (e - rx).$$

Since $x \in J(eRe)$, then there exists $t = er_1e \in eRe$ such that

$$t(e - ex) = e.$$

Hence we have

$$t(e - rx) = er_1e^2(e - rx) = t(e - ex) = e.$$

Now we consider

$$(1 - e + t)(1 - rx) = (1 - e + t)(1 - e + e - rx) = 1 - (1 - e)rx$$

Hence we have

$$((1 - e)rx)^2 = (1 - e)rx(1 - e)rx = 0$$

since $x \in eRe$ and $x(1 - e) = 0$. Let $\varphi = (1 - e + t)$ and $\psi = (1 - rx)$ we then have

$$(2\varphi - \varphi\psi\varphi)(1 - rx) = 1.$$

It means that $x \in J(R)$. Equivalently, we prove that $J(eRe) \subseteq J(R)$.

Since $J(eRe) \subseteq eRe$ we obtain that

$$J(eRe) \subseteq eRe \cap J(R).$$

Let $x \in eRe \cap J(R)$, then we can suppose that $x = er_0e$ and for any $r \in R$, we have

$$1 - rx = 1 - r(er_0e)$$

is invertible. Suppose that $t(1 - rx) = 1$ with $r = er'e$ then we consider

$$(ete)(e - rx) = (ete)(e - (er'e)(er_0e)) = et(1 - rx)e = e^2 = e.$$

it implies that

$$J(eRe) \supseteq eRe \cap J(R).$$

Then we obtain that $J(eRe) = eRe \cap J(R)$.

Next, we consider $eJ(R)e$. we consider that R as a left R -module ${}_R R$. Then we have idempotent decomposition

$${}_R R = eR \oplus (1 - e)R$$

Then we have $J(R) = eJ(R) + (1 - e)J(R)$ and

$$J(R) \cap eR = eJ(R). \quad (\star)$$

Similarly, if we consider R as a right R -module R_R , we obtain that

$$J(R) \cap Re = J(R)e. \quad (\Delta)$$

I claim that $eR \cap Re = eRe$. To see that, let $x \in eR \cap Re$ then suppose that $x = er_1 = r_2e$. Since e is idempotent, then we have

$$er_1e = xe = r_2e^2 = r_2e = x$$

it implies that $eR \cap Re \subseteq eRe$. And $eRe \subseteq eR \cap Re$ is easy to see. By the same reason we have $eJ(R) \cap J(R)e = eJ(R)e$. Then we obtain that

$$(J(R) \cap eR) \cap (J(R) \cap Re) = J(R) \cap eRe$$

and therefore $J(R) \cap eRe = eJ(R)e$.

(ii) Let E_{ij} denote the matrix of (i, j) -entry is 1_R and others entires is zero. Let M denote $M_n(R)$. Let $A = (a_{ij})$ be an arbitrary element in $J(M)$. Since $J(M)$ is a double sided ideal, then we have $\tilde{A} = E_{1i}AE_{j1} \in J(M)$. According to (i), we have

$$E_{11}J(M)E_{11} = J(E_{11}ME_{11}) \simeq J(R).$$

Then it implies that $E_{11}\tilde{A}E_{11} \in J(E_{11}ME_{11})$. Equivalently, we have $a_{ij} \in J(R)$ and thus $J(M) \subseteq M_n(J(R))$. Conversely, we first show that $J(R)E_{ii} \subseteq J(M)$ since

$$J(R)E_{ii} = J(E_{ii}ME_{ii}) = E_{ii}J(M)E_{ii} \subseteq J(M).$$

Therefore, we have

$$J(R)E_{ij} = J(R)E_{ii}E_{ij} \subseteq J(M)E_{ij} \subseteq J(M).$$

It implies that $M_n(J(R)) \subseteq J(M)$ and thus

$$J(M_n(R)) = M_n(J(R)).$$

■

Problem 3

Prove: A submodule of a finitely generated free module over a PID is free.

Solution: Let R be an arbitrary PID. We first to show that submodules of ${}_R R$ is free. Let H be an arbitrary submodule of R , equivalently, H be an arbitrary ideal of R . Since R is PID, then there exists an element $x \in R$ such that $H = Rx$. Since R is integral then there is not nontrivial zero divisor in R . Therefore, we find that

$$R \rightarrow Rx \quad r \mapsto rx$$

is an isomorphism. Equivalently, H is free R -module.

Now, we consider the induction on rank of finitely generated free modules. Suppose that our claim holds for R^{n-1} . We consider the case that rank of n . Let H be an arbitrary submodule of R^n . Let p_n be the canonical projection on n^{th} position of R^n . Then we obtain a short exact sequence

$$0 \rightarrow \ker p_n \cap H \rightarrow H \rightarrow p_n(H) \rightarrow 0.$$

Since $p_n(H)$ is a submodule of R , it means that $p_n(H)$ is a free R -module. Therefore, the exact sequence split. We can write

$$H \simeq (\ker p_n \cap H) \oplus p_n(H).$$

Since $\ker p_n \cap H$ is the submodule of R^{n-1} , then by our assumption, it is also a free R -module. Therefore, our claim holds. ■

14 Week 11

Problem 1

Let R be a ring and V be an R -module. Let I be an ideal of R . Suppose that d_I is defined on V and on R . Then

- (i) Addition is continuous from $V \times V$ to V .
- (ii) Multiplication is continuous from $R \times V$ to V .
- (iii) If d_I is defined on the R -module W and $f \in \text{Hom}_R(V, W)$, then f is continuous.

Solution: (1) Let the addition map be $f : V \times V \rightarrow V$, where $f(x, y) = x + y$. Let O be an arbitrary open set in the target space V . We need to prove that its preimage $f^{-1}(O)$ is an open set in $V \times V$. If $f^{-1}(O)$ is empty, it is trivially open. If $f^{-1}(O)$ is non-empty, pick an arbitrary point $(x, y) \in f^{-1}(O)$. It means $x + y \in O$. Since O is an open set in V , there exists an integer n such that:

$$(x + y) + I^n V \subseteq O.$$

Let $U_x = x + I^n V$ and $U_y = y + I^n V$ be the open neighborhood of x, y respectively. Then $U_x \times U_y$ is an open neighborhood of (x, y) in $V \times V$. Since

$$f(U_x \times U_y) = U_x + U_y = (x + I^n V) + (y + I^n V) = x + y + I^n V$$

we have $f(U_x \times U_y) \subseteq O$ which implies

$$U_x \times U_y \subseteq f^{-1}(O).$$

Therefore, for arbitrary $(x, y) \in f^{-1}(O)$, we have an open neighborhood $U_x \times U_y$ contained in $f^{-1}(O)$. It turns out that $f^{-1}(O)$ is open and thus f is continuous.

(2) Let the multiplication map be $g : R \times V \rightarrow V$, where $g(r, v) = rv$. Let O be an arbitrary open set in the target space V . We need to prove that its preimage $g^{-1}(O)$ is an open set in $R \times V$. If $g^{-1}(O)$ is empty, it is trivially open. If $g^{-1}(O)$ is non-empty, pick an arbitrary point $(r, v) \in g^{-1}(O)$. It means $rv \in O$. Since O is an open set in V , there exists an integer n such that:

$$rv + I^n V \subseteq O.$$

Let $U_r = r + I^n$ and $U_v = v + I^n V$ be the open neighborhood of r, v respectively. Then $U_r \times U_v$ is an open neighborhood of (r, v) in $R \times V$. Since

$$g(U_r \times U_v) = U_r U_v = (r + I^n)(v + I^n V) \subseteq rv + r(I^n V) + (I^n)v + I^{2n}V \subseteq rv + I^n V$$

we have $g(U_r \times U_v) \subseteq O$ which implies

$$U_r \times U_v \subseteq g^{-1}(O).$$

Therefore, for arbitrary $(r, v) \in g^{-1}(O)$, we have an open neighborhood $U_r \times U_v$ contained in $g^{-1}(O)$. It turns out that $g^{-1}(O)$ is open and thus g is continuous.

(3) Let O be an arbitrary open set in the target space W . We need to prove that its preimage $f^{-1}(O)$ is an open set in V . If $f^{-1}(O)$ is empty, it is trivially open. If $f^{-1}(O)$ is non-empty, pick an arbitrary point $v \in f^{-1}(O)$. It means $f(v) \in O$. Since O is an open set in W , there exists an integer n such that:

$$f(v) + I^n W \subseteq O.$$

Let $U_v = v + I^n V$ be the open neighborhood of v in V . Since f is an R -module homomorphism, we have $f(I^n V) = I^n f(V) \subseteq I^n W$. Then

$$f(U_v) = f(v + I^n V) = f(v) + f(I^n V) \subseteq f(v) + I^n W$$

we have $f(U_v) \subseteq O$ which implies

$$U_v \subseteq f^{-1}(O).$$

Therefore, for arbitrary $v \in f^{-1}(O)$, we have an open neighborhood U_v contained in $f^{-1}(O)$. It turns out that $f^{-1}(O)$ is open and thus f is continuous. ■

Problem 2

The following algebra isomorphisms hold for R -algebras A and B , where R is a commutative ring.

$$(i) \quad M_n(A) \otimes_R B \simeq M_n(A \otimes_R B).$$

$$(ii) \quad M_n(A) \otimes_R M_m(B) \simeq M_{nm}(A \otimes_R B).$$

Solution: (i) Let

$$\begin{aligned} \phi : M_n(A) \times B &\rightarrow M_n(A \otimes_R B) \\ ((a_{ij}), b) &\mapsto (a_{ij} \otimes b) \end{aligned}$$

Since ϕ is R -bilinear, it induces a well-defined R -linear map

$$\begin{aligned} \Phi : M_n(A) \otimes_R B &\rightarrow M_n(A \otimes_R B) \\ (a_{ij}) \otimes b &\mapsto (a_{ij} \otimes b) \end{aligned}$$

For any $(a_{ij}), (c_{ij}) \in M_n(A)$ and $b, d \in B$, we have

$$\Phi(((a_{ij}) \otimes b)((c_{ij}) \otimes d)) = \Phi((a_{ij})(c_{ij}) \otimes bd) = \left(\sum_k a_{ik} c_{kj} \otimes bd \right).$$

On the other hand,

$$\Phi((a_{ij}) \otimes b) \Phi((c_{ij}) \otimes d) = (a_{ij} \otimes b)(c_{ij} \otimes d) = \left(\sum_k (a_{ik} \otimes b)(c_{kj} \otimes d) \right) = \left(\sum_k a_{ik} c_{kj} \otimes bd \right).$$

Thus Φ preserves multiplication and is an algebra homomorphism. The map

$$\begin{aligned} \Psi : M_n(A \otimes_R B) &\rightarrow M_n(A) \otimes_R B \\ (a \otimes b)E_{ij} &\mapsto aE_{ij} \otimes b \end{aligned}$$

provides a two-sided inverse to Φ . Therefore, we obtain that

$$M_n(A) \otimes_R B \simeq M_n(A \otimes_R B).$$

(ii) By applying the isomorphism from (i) and treating $M_m(B)$ as an R -algebra, we have

$$M_n(A) \otimes_R M_m(B) \simeq M_n(A \otimes_R M_m(B)).$$

Since R is commutative, the tensor product is symmetric up to isomorphism, yielding

$$A \otimes_R M_m(B) \simeq M_m(B) \otimes_R A.$$

Applying (i) again to $M_m(B) \otimes_R A$, we get

$$M_m(B \otimes_R A) \simeq M_m(A \otimes_R B).$$

Substituting this sequence of isomorphisms back gives:

$$M_n(A \otimes_R M_m(B)) \simeq M_n(M_m(A \otimes_R B)).$$

For any ring S , there is a natural ring isomorphism $M_n(M_m(S)) \simeq M_{nm}(S)$ given by viewing an $n \times n$ block matrix of $m \times m$ matrices as an $nm \times nm$ matrix. Letting $S = A \otimes_R B$, we conclude

$$M_n(A) \otimes_R M_m(B) \simeq M_{nm}(A \otimes_R B).$$

■

15 Week 12

Problem 1

- (i) Let (X, ρ) be a transitive **Set**-representation of G . For $x \in X$, let G_x denote the fixator of x in G . Then the map

$$G/G_x \rightarrow X, \quad gG_x \mapsto \rho(g)(x)$$

defines an isomorphism $(G/G_x, \rho_{G_x}) \xrightarrow{\sim} (X, \rho)$.

- (ii) For subgroups H and H' of G , the representations $(G/H, \rho_H)$ and $(G/H', \rho_{H'})$ are isomorphic if and only if H and H' are conjugate in G .

Solution: (i) Let $f : G/G_x \rightarrow X$ denote the map we discuss. First we show that f is well-defined. Suppose

$$gG_x = hG_x.$$

Then $h^{-1}g \in G_x$, so

$$\rho_{h^{-1}g}(x) = x.$$

Hence

$$\rho_g(x) = \rho_h(\rho_{h^{-1}g}(x)) = \rho_h(x).$$

Therefore $f(gG_x) = f(hG_x)$, so f is well-defined. Next, We show that f is a bijection of sets. Let y be an arbitrary element of X that distinct to x . Since G act transitively on X then there exists $g \in G$ such that $\rho_g(x) = y$. Therefore, we have

$$f(gG_x) = \rho_g(x) = y.$$

It implies that f is surjective. Let $g, h \in G$ such that $f(gG_x) = f(hG_x)$. It implies that $\rho_g(x) = \rho_h(x)$ and thus

$$x = \rho_{h^{-1}} \circ \rho_g(x) = \rho_{h^{-1} \cdot g}(x).$$

Then we have $h^{-1} \cdot g \in G_x$. Equivalently, we have $gG_x = hG_x$. Therefore, f is injective. For arbitrary $g \in G$, we then have

$$f(g \cdot hG_x) = f(ghG_x) = \rho_{gh}(x) = \rho_g \circ \rho_h(x) = \rho_g \cdot f(hG_x).$$

Therefore, f is isomorphism of representation.

(ii) Suppose first that

$$f : G/H \rightarrow G/H'$$

is an isomorphism of G -sets. Write

$$f(H) = gH'$$

for some $g \in G$. Since f is G -equivariant, it preserves stabilizers. Hence

$$H = \text{Stab}_G(H) = \text{Stab}_G(f(H)) = \text{Stab}_G(gH') = gH'g^{-1}.$$

Thus H and H' are conjugate.

Conversely, suppose

$$H = gH'g^{-1}$$

for some $g \in G$. Define

$$\Phi : G/H \rightarrow G/H', \quad \Phi(aH) = agH'.$$

This is well-defined: if $aH = bH$, then $a = bh$ for some $h \in H$. Since

$$g^{-1}hg \in H',$$

we get

$$agH' = bhgH' = bgH'.$$

Thus $\Phi(aH) = \Phi(bH)$.

The map Φ is bijective, with inverse

$$\Psi : G/H' \rightarrow G/H, \quad \Psi(aH') = ag^{-1}H.$$

Moreover, for $k \in G$,

$$\Phi(k \cdot aH) = \Phi(kaH) = kagH' = k \cdot \Phi(aH).$$

Therefore Φ is G -equivariant. Hence

$$(G/H, \rho_H) \cong (G/H', \rho_{H'})$$

if and only if H and H' are conjugate in G . ■

Problem 2

Assume that G is the cyclic (additive) group $\mathbb{Z}/p\mathbb{Z}$ where p is a prime number, and let k be a field of characteristic p . Consider the morphism

$$G \rightarrow \mathrm{GL}_2(k), \quad \lambda \mapsto \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}.$$

Prove that the line generated by the first standard basis vector of k^2 is G -stable but has no G -stable complement.

Solution: Let

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Since $\mathrm{char}(k) = p$, the map

$$\rho : \mathbb{Z}/p\mathbb{Z} \rightarrow \mathrm{GL}_2(k), \quad \lambda \mapsto \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}$$

is well-defined and is a group homomorphism.

Let

$$W = ke_1.$$

Then for every $\lambda \in \mathbb{Z}/p\mathbb{Z}$,

$$\rho(\lambda)e_1 = e_1,$$

so W is G -stable.

We claim that W has no G -stable complement. Suppose, for contradiction, that

$$k^2 = W \oplus U$$

with U is G -stable. Then U is one-dimensional and $U \neq W$, so

$$U = k(ae_1 + e_2)$$

for some $a \in k$. Since U is G -stable, it is stable under the action of $1 \in \mathbb{Z}/p\mathbb{Z}$. Thus

$$\rho(1)(ae_1 + e_2) = (a+1)e_1 + e_2$$

must lie in U . Hence there exists $c \in k$ such that

$$(a+1)e_1 + e_2 = c(ae_1 + e_2).$$

Comparing coefficients gives $c = 1$, and then

$$a+1 = a,$$

a contradiction. Therefore W has no G -stable complement. ■

Problem 3

Let (M, ρ) and (N, σ) be two k -representations of G . Fix basis $(e_i)_{1 \leq i \leq m}$ and $(f_j)_{1 \leq j \leq n}$ of M and N respectively. Then $(e_i \otimes f_j)_{1 \leq i \leq m, 1 \leq j \leq n}$ is a basis of $M \otimes_k N$. Denote by (m, ρ) , (n, σ) , $(mn, \rho \otimes \sigma)$ be the matrix representations associated with (M, ρ) , (N, σ) , $(M \otimes_k N, \rho \otimes \sigma)$ respectively. Determine the relation between $(\rho \otimes \sigma)(g)$ and $\rho(g)$, $\sigma(g)$.

Solution: Suppose

$$\rho_g(e_i) = \sum_{k=1}^m a_{ki} e_k, \quad \sigma_g(f_j) = \sum_{\ell=1}^n b_{\ell j} f_\ell.$$

Then

$$(\rho \otimes \sigma)_g(e_i \otimes f_j) = \rho_g(e_i) \otimes \sigma_g(f_j) = \sum_{k=1}^m \sum_{\ell=1}^n a_{ki} b_{\ell j} (e_k \otimes f_\ell).$$

With respect to the ordered basis

$$(e_1 \otimes f_1, \dots, e_1 \otimes f_n, e_2 \otimes f_1, \dots, e_m \otimes f_n),$$

the column corresponding to $e_i \otimes f_j$ is

$$(a_{1i} b_{1j}, \dots, a_{1i} b_{nj}, a_{2i} b_{1j}, \dots, a_{2i} b_{nj}, \dots, a_{mi} b_{1j}, \dots, a_{mi} b_{nj})^T.$$

Therefore

$$[(\rho \otimes \sigma)_g] = [\rho_g] \otimes [\sigma_g],$$

where the tensor product on the right is the Kronecker product of matrices. ■

Problem 4

Let (M, ρ) be a k -representation of G . Fix basis $(e_i)_{1 \leq i \leq m}$ of M . Let (M^*, ρ^*) be the contragredient representation of (M, ρ) . Denote by (m, ρ) , (m, ρ^*) be the matrix representations associated with (M, ρ) , (M^*, ρ^*) . Determine the relation between $\rho(g)$ and $\rho^*(g)$.

Solution: Let $e_1, \dots, e_n$ be a basis of V , with dual basis $e_1^*, \dots, e_n^*$. Suppose

$$\rho_g(e_i) = \sum_{k=1}^n a_{ki} e_k.$$

Thus

$$[\rho_g] = (a_{ki}).$$

The contragredient representation ρ^* is defined by

$$(\rho_g^* \varphi)(v) = \varphi(\rho_{g^{-1}} v).$$

Write

$$\rho_{g^{-1}}(e_j) = \sum_{k=1}^n b_{kj} e_k.$$

Then $[\rho_{g^{-1}}] = (b_{kj}) = [\rho_g]^{-1}$. For each i, j ,

$$(\rho_g^* e_i^*)(e_j) = e_i^*(\rho_{g^{-1}} e_j) = e_i^* \left(\sum_{k=1}^n b_{kj} e_k \right) = b_{ij}.$$

Hence the coefficient of e_j^* in $\rho_g^*(e_i^*)$ is b_{ij} . Therefore

$$[\rho_g^*] = (b_{ij}) = [\rho_{g^{-1}}]^T = ([\rho_g]^{-1})^T.$$

Thus the matrix of the contragredient representation is the inverse transpose of the original matrix. ■

Problem 5

- (i) Let G be a finite group acting on a finite set Ω . Let $\text{Fix}^G(\Omega)$ denote the set of fixed points of Ω under G , and let $\Omega^\#$ denote the set of orbits of G on $\Omega \setminus \text{Fix}^G(\Omega)$. Prove that

$$|\Omega| = |\text{Fix}^G(\Omega)| + \sum_{\omega \in \Omega^\#} |\omega|.$$

From now on, we assume that p is a prime number and that G is a nontrivial p -group.

- (ii) Assume now that $|\Omega|$ is a power of p different from 1. Prove that $|\text{Fix}^G(\Omega)|$ is divisible by p .
- (iii) Let k be a (commutative) field of characteristic p . Let M be a kG -module. Prove that $\text{Fix}^G(M) \neq 0$. (Hint. (a) Let $(e_i)_{i=1, \dots, r}$ be a k -basis of M . Prove that the abelian group generated by $(ge_i)_{i=1, \dots, r} (g \in G)$ is an $\mathbb{F}_p G$ -module. (b) Apply question (2) above to conclude.)
- (iv) Prove that, up to isomorphism, the trivial representation of G is the unique irreducible kG -module.

Solution: (i) The set Ω decomposes as a disjoint union

$$\Omega = \text{Fix}^G(\Omega) \sqcup (\Omega \setminus \text{Fix}^G(\Omega)).$$

Moreover, $\Omega \setminus \text{Fix}^G(\Omega)$ is G -stable, hence it is a disjoint union of its G -orbits:

$$\Omega \setminus \text{Fix}^G(\Omega) = \bigsqcup_{\omega \in \Omega^\#} \omega.$$

Therefore

$$\Omega = \text{Fix}^G(\Omega) \sqcup \bigsqcup_{\omega \in \Omega^\#} \omega.$$

Taking cardinalities gives

$$|\Omega| = |\text{Fix}^G(\Omega)| + \sum_{\omega \in \Omega^\#} |\omega|.$$

(ii) We just prove that for any $\omega \in \Omega^\#$ we have $|\omega|$ is divisible by p . Consider ω as transitive G -set, according to (1), we choose $x \in \omega$ then it induces the bijection

$$G/G_x \xrightarrow{\sim} \omega.$$

Since the only prime divisor of $|G|$ is p , we can suppose that $|\omega| = |G/G_x| = p^s$. If $s = 0$, it turns out that $x \in \text{Fix}^G(\Omega)$, which contradicts to our assumption. Therefore, $|\Omega|$ and each $|\omega|$ is divisible by p implies that

$$|\text{Fix}^G(\Omega)| = |\Omega| - \sum_{\omega \in \Omega^\#} |\omega|.$$

is also divisible by p .

- (iii) Assume $M \neq 0$, and let $e_1, \dots, e_r$ be a k -basis of M . Set

$$M' = \langle ge_i \mid g \in G, 1 \leq i \leq r \rangle_{\mathbb{F}_p} \subset M.$$

Since $\text{char}(k) = p$, this is a finite $\mathbb{F}_p$ -subspace of M . It is G -stable, because for $h \in G$,

$$h(ge_i) = (hg)e_i \in M'.$$

Thus G acts on the finite set M' .

Since $M' \neq 0$, we have $|M'| = p^n$ for some $n \geq 1$. By part (ii),

$$|\text{Fix}^G(M')|$$

is divisible by p . But $0 \in \text{Fix}^G(M')$, so the fixed point set cannot consist only of 0. Hence there exists

$$0 \neq m \in \text{Fix}^G(M').$$

Since $M' \subset M$, we get

$$0 \neq m \in \text{Fix}^G(M).$$

Therefore

$$\text{Fix}^G(M) \neq 0.$$

(iv) Let M be an irreducible kG -module. By part (iii),

$$\text{Fix}^G(M) \neq 0.$$

Choose

$$0 \neq m \in \text{Fix}^G(M).$$

Then $gm = m$ for all $g \in G$. Hence the one-dimensional subspace

$$km \subset M$$

is G -stable. Since M is irreducible, we must have

$$M = km.$$

Thus M is one-dimensional and G acts trivially on M . Therefore

$$M \cong k$$

as kG -modules, where k is the trivial representation.

Conversely, the trivial representation k is one-dimensional, hence irreducible. Therefore, up to isomorphism, the trivial representation is the unique irreducible kG -module. ■

16 Week 13

Problem 1

Write out a proof of Maschke's Theorem in the case of representations over $\mathbb{C}$ along the following lines.

1. Given a representation

$$\rho : G \rightarrow GL(V),$$

where V is a vector space over $\mathbb{C}$, let $(\cdot, \cdot)$ be any positive definite Hermitian form on V . Define a new form $(\cdot, \cdot)_1$ on V by

$$(v, w)_1 = \frac{1}{|G|} \sum_{g \in G} (gv, gw).$$

Show that $(\cdot, \cdot)_1$ is a positive definite Hermitian form, preserved under the action of G ; that is,

$$(v, w)_1 = (gv, gw)_1$$

always.

2. If W is a subrepresentation of V , show that

$$V = W \oplus W^\perp$$

as representations.

Solution: (1) For $a, b \in \mathbb{C}$,

$$(av_1 + bv_2, w)_1 = \frac{1}{|G|} \sum_{g \in G} (a gv_1 + b gv_2, gw) = a(v_1, w)_1 + b(v_2, w)_1,$$

and similarly the second variable is conjugate-linear. Moreover,

$$(w, v)_1 = \frac{1}{|G|} \sum_{g \in G} (gw, gv) = \overline{\frac{1}{|G|} \sum_{g \in G} (gv, gw)} = \overline{(v, w)_1}.$$

Thus $(\cdot, \cdot)_1$ is Hermitian. Next, for $v \neq 0$, since each $g \in G$ acts invertibly, we have $gv \neq 0$. Hence

$$(v, v)_1 = \frac{1}{|G|} \sum_{g \in G} (gv, gv) > 0.$$

Therefore $(\cdot, \cdot)_1$ is positive definite. Finally, we show that this form is preserved by the action of G . Let $h \in G$. Then

$$(hv, hw)_1 = \frac{1}{|G|} \sum_{g \in G} (ghv, ghw).$$

As g runs over G , so does gh . Therefore

$$(hv, hw)_1 = \frac{1}{|G|} \sum_{u \in G} (uv, uw) = (v, w)_1.$$

Thus

$$(v, w)_1 = (hv, hw)_1$$

for all $h \in G$, so $(\cdot, \cdot)_1$ is G -invariant.

(2) Choose a basis of V , so that $V \simeq \mathbb{C}^n$, and take the standard positive definite Hermitian form

$$(v, w) = \sum_i v_i \overline{w_i}.$$

Averaging this form over the G -action gives a new Hermitian form

$$(v, w)_1 = \frac{1}{|G|} \sum_{g \in G} (gv, gw).$$

By the previous part, $(\cdot, \cdot)_1$ is positive definite and G -invariant.

Now let $W \subset V$ be a subrepresentation, and define

$$W^\perp = \{v \in V \mid (v, w)_1 = 0 \text{ for all } w \in W\}.$$

Since $(\cdot, \cdot)_1$ is positive definite, we have

$$V = W \oplus W^\perp$$

as vector spaces.

It remains to show that $W^\perp$ is G -stable. Let $u \in W^\perp$, $h \in G$, and $w \in W$. Since W is a subrepresentation, $h^{-1}w \in W$. Using G -invariance of $(\cdot, \cdot)_1$, we get

$$(hu, w)_1 = (u, h^{-1}w)_1 = 0.$$

Hence $hu \in W^\perp$, so $W^\perp$ is a subrepresentation. Therefore

$$V = W \oplus W^\perp$$

as representations. ■

Problem 2

Prove the following theorem of Burnside: let G be a finite group, k an algebraically closed field in which $|G|$ is invertible, and

$$\rho : G \rightarrow GL(V)$$

be a representation over k . By taking a basis of V , write each endomorphism $\rho(g)$ as a matrix. Let $\dim V = n$. Show that the representation is simple if and only if there exist n^2 elements $g_1, \dots, g_{n^2}$ of G so that the matrices

$$\rho(g_1), \dots, \rho(g_{n^2})$$

are linearly independent, and that this happens if and only if the algebra homomorphism

$$kG \rightarrow \text{End}_k(V)$$

is surjective. Note that ρ itself is generally not surjective.

Solution: Let

$$A = \text{span}_k\{\rho(g) : g \in G\} \subseteq \text{End}_k(V).$$

Then A is the image of the algebra homomorphism $kG \rightarrow \text{End}_k(V)$.

Suppose that there exist $g_1, \dots, g_{n^2} \in G$ such that $\rho(g_1), \dots, \rho(g_{n^2})$ are linearly independent. Since $\dim_k \text{End}_k(V) = n^2$, they form a basis of $\text{End}_k(V)$. Hence every elementary matrix E_{ij} , sending e_j to e_i and all other basis vectors to 0, can be written as a k -linear combination of the matrices $\rho(g)$. Let $0 \neq W \subseteq V$ be a subrepresentation and choose $0 \neq v = \sum_{\ell=1}^n c_\ell e_\ell \in W$. Pick j with $c_j \neq 0$. Then for every i ,

$$\frac{1}{c_j} E_{ij} v = e_i.$$

Since W is stable under every $\rho(g)$, it is stable under their linear combinations. Thus $e_i \in W$ for all i , so $W = V$. Therefore V is simple.

Conversely, suppose that V is simple. Let

$$A = \text{im}(kG \rightarrow \text{End}_k(V)).$$

Since $|G|$ is invertible in k , Maschke's theorem implies that kG is semisimple. Hence its quotient A is also semisimple.

Since V is simple as a G -representation, it is simple as an A -module. By Schur's lemma, because k is algebraically closed,

$$\text{End}_A(V) = k.$$

By the Artin–Wedderburn theorem, A is a product of matrix algebras over k . Since V is simple, only one simple factor acts nontrivially on V . But A is defined as the image in $\text{End}_k(V)$, so its action on V is faithful. Therefore A has only this one factor, and its action on V identifies it with

$$\text{End}_k(V).$$

Thus

$$A = \text{End}_k(V).$$

Hence $\dim_k A = n^2$. Since A is spanned by the matrices $\rho(g)$, one can choose n^2 linearly independent matrices among them. ■

Problem 3

Let

$$G = \langle x \mid x^n = 1 \rangle$$

be a cyclic group of order n , and let $\zeta_n \in \mathbb{C}$ be a primitive n -th root of unity. Then the simple complex characters of G are the n functions

$$\chi_r(x^s) = \zeta_n^{rs},$$

where $0 \leq r \leq n-1$.

Solution: Let V be a simple complex representation of G , and set $T = \rho(x)$. Since $x^n = 1$, we have $T^n = I$. Hence the minimal polynomial of T divides $t^n - 1$. Over $\mathbb{C}$, the polynomial $t^n - 1 = \prod_{r=0}^{n-1} (t - \zeta_n^r)$ has distinct roots, so T is diagonalizable and its eigenvalues are among $\zeta_n^0, \dots, \zeta_n^{n-1}$.

For an eigenvalue λ , the eigenspace $V_\lambda = \{v \in V \mid Tv = \lambda v\}$ is G -stable, because $G = \langle x \rangle$. Since V is simple, only one eigenspace can be nonzero. Thus $V = V_\lambda$ for some λ , so $\dim V = 1$.

Therefore every simple complex representation of G is one-dimensional. Since $\rho(x)^n = 1$, we have $\rho(x) = \zeta_n^r$ for some $0 \leq r \leq n-1$. Hence, for every s ,

$$\chi_r(x^s) = \rho(x^s) = \rho(x)^s = (\zeta_n^r)^s = \zeta_n^{rs}.$$

These n characters are distinct since $\chi_r(x) = \zeta_n^r$. Hence they are exactly all the simple complex characters of G . ■

17 Week 14

Problem 1

Let G be a finite group and let p be a prime. Prove that each element $x \in G$ can be uniquely written as

$$x = st,$$

where s is p -regular, t is p -singular, and $st = ts$. If $x_1 = s_1 t_1$ is such a decomposition of an element x_1 that is conjugate to x , then s is conjugate to s_1 , and t is conjugate to t_1 .

Solution: Let $x \in G$, and write $|x| = n = p^a m$, where $(m, p) = 1$. Choose integers A, B such that $Ap^a + Bm = 1$. Set $s = x^{Ap^a}$ and $t = x^{Bm}$. Then s and t commute, since both are powers of x , and $st = x^{Ap^a+Bm} = x$. Moreover, $s^m = x^{Ap^a m} = x^{An} = 1$, so $|s| \mid m$; hence s is p -regular. Similarly, $t^{p^a} = x^{Bmp^a} = x^{Bn} = 1$, so $|t| \mid p^a$; hence t is a p -element. Thus the required decomposition exists.

Now suppose $x = s_1 t_1$ is another such decomposition, with s_1 p -regular, t_1 a p -element, and $s_1 t_1 = t_1 s_1$. Since s_1 and t_1 commute and have relatively prime orders, $|x| = |s_1 t_1| = |s_1| |t_1|$. Comparing the p -part and the prime-to- p part of $|x| = p^a m$, we get $|s_1| = m$ and $|t_1| = p^a$. Therefore

$$x^{Ap^a} = (s_1 t_1)^{Ap^a} = s_1^{Ap^a} t_1^{Ap^a} = s_1,$$

because $Ap^a \equiv 1 \pmod{m}$ and $t_1^{p^a} = 1$. Similarly, $x^{Bm} = (s_1 t_1)^{Bm} = t_1$. Hence $s_1 = s$ and $t_1 = t$, so the decomposition is unique.

Finally, if $x_1 = g x g^{-1}$, then $x_1 = (g s g^{-1})(g t g^{-1})$. The first factor is still p -regular, the second is still a p -element, and they commute. By uniqueness, if $x_1 = s_1 t_1$ is the corresponding decomposition, then $s_1 = g s g^{-1}$ and $t_1 = g t g^{-1}$. Thus s is conjugate to s_1 , and t is conjugate to t_1 . ■